MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INTEGRATOR, S/D, ANALOG, KEY CODE PEG

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic S/D integrator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.1285 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be PEG.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 100 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A21 through A40. Additional contact pins A1 through A20 and B21 through B40 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A and a heat dissipation fin on increment B.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/228-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1, 2, and 4 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for power supply voltage (V_{CC}) of 5.0 V dc ±10 percent and ±15 V dc ±10 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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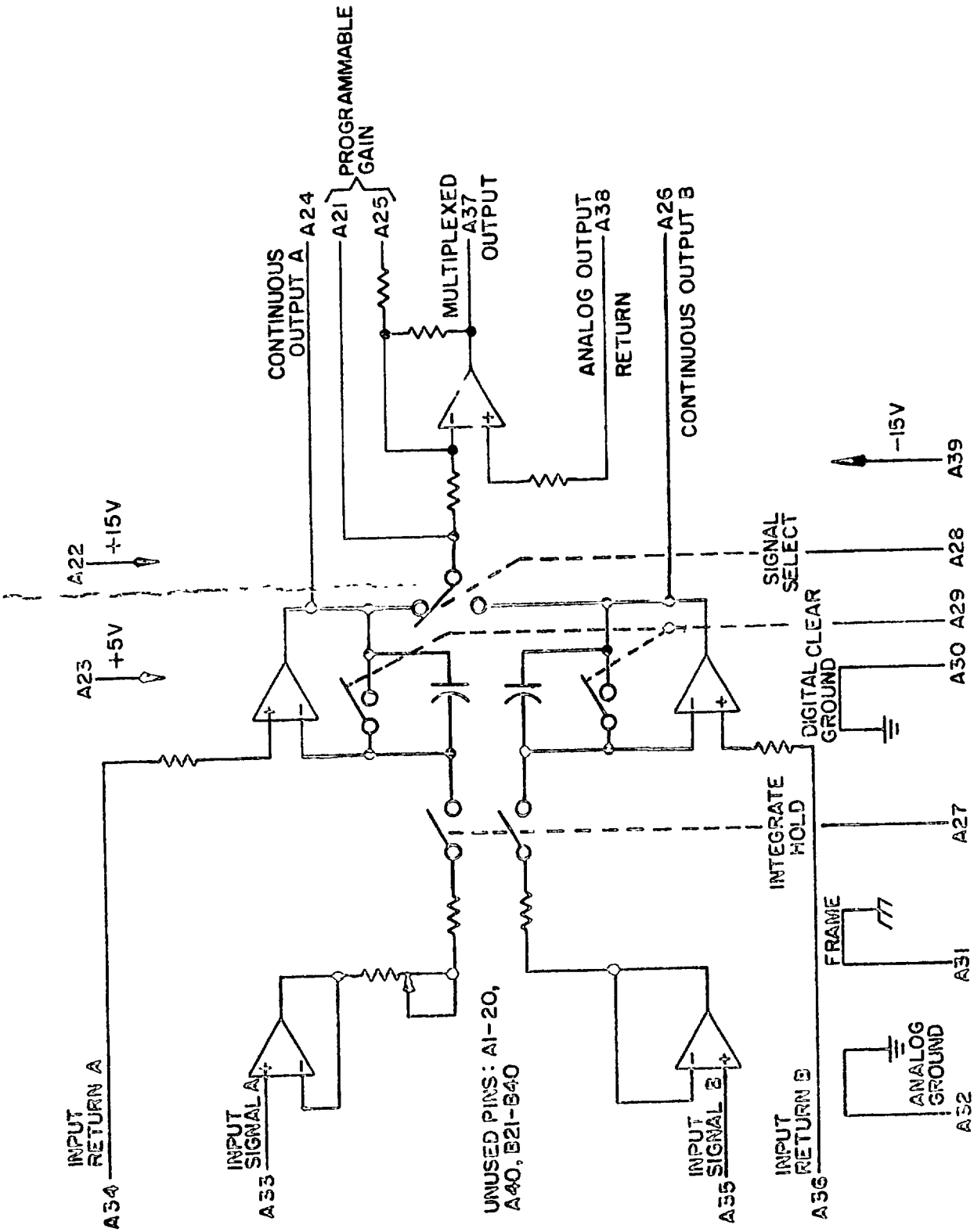


FIGURE 1. PEG module function diagram, S/D integrator.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Maximum stress	II	S	5	60	5	60	5	60	5	60	s
B1-B3	Isolation	III	X	10	∞	1	∞	10	∞	1	∞	MΩ
C1-C3	Input breakdown current	IV	X	0	1	0	1	0	1	0	1	mA
D1-D4	Ground return isolation	V	S	10	∞	10	∞	10	∞	10	∞	kΩ
E	Frame continuity	VI	S	0	1	0	1	0	1	0	1	Ω
F1-F10	Voltage sequencing	VII	S	5	60	5	60	5	60	5	60	s
G1	DC offset voltage at 280 µs	VIII	X	-1.8	1.8	-2.0	2.0	-2.0	2.0	-2.5	2.5	mV
G2	DC offset voltage at 150 µs	VIII	X	-1.8	1.8	-2.0	2.0	-2.0	2.0	-2.5	2.5	mV
G3	DC gain check at 280 µs	VIII	X	5.60	5.73	5.55	5.74	5.55	5.74	5.53	5.74	V
G4	DC gain check at 150 µs	VIII	X	-1.5	1.5	-6.0	6.0	-6.0	6.0	-8.0	8.0	mV
H1	Angular accuracy check (-sin 0°)	IX	X	-3.0	3.0	-4.0	4.0	-4.0	4.0	-5.0	5.0	mV
H2	Angular accuracy check (-cos 0°)	IX	X	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V
H3	Angular accuracy check (-sin 90°)	IX	X	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V
H4	Angular accuracy check (-cos 90°)	IX	X	-3.0	3.0	-4.0	4.0	-4.0	4.0	-5.0	5.0	mV
H5	Angular accuracy check (-sin 180°)	IX	X	-3.0	3.0	-4.0	4.0	-4.0	4.0	-5.0	5.0	mV
H6	Angular accuracy check (-cos 180°)	IX	X	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
H7	Angular accuracy check (-sin 270°)	IX	X	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M8	Angular accuracy check (-cos 270°)	IX	X	-3.0	3.0	-4.0	4.0	-4.0	4.0	-5.0	5.0	mV
M9	Angular accuracy check (sin 45° -cos 45°)	IX	X	-4.2	4.2	-5.6	5.6	-5.6	5.6	-7.0	7.0	mV
M10	Angular accuracy check (sin 135° +cos 135°)	IX	X	-2.1	2.1	-2.8	2.8	-2.8	2.8	-3.5	3.5	mV
M11	Angular accuracy check (sin 225° -cos 225°)	IX	X	-4.2	4.2	-5.6	5.6	-5.6	5.6	-7.0	7.0	mV
M12	Angular accuracy check (sin 315° +cos 315°)	IX	X	-2.1	2.1	-2.8	2.8	-2.8	2.8	-3.5	3.5	mV
J1	Angular accuracy check (sin 0°)	X	X	-2.2	2.2	-2.9	2.9	-2.9	2.9	-3.6	3.6	mV
J2	Angular accuracy check (cos 0°)	X	X	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
J3	Angular accuracy check (sin 90°)	X	X	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
J4	Angular accuracy check (cos 90°)	X	X	-2.2	2.2	-2.9	2.9	-2.9	2.9	-3.6	3.6	mV
J5	Angular accuracy check (sin 180°)	X	X	-2.2	2.2	-2.9	2.9	-2.9	2.9	-3.6	3.6	mV
J6	Angular accuracy check (cos 180°)	X	X	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V
J7	Angular accuracy check (sin 270°)	X	X	-3.18	-3.73	-3.18	03.73	-3.18	-3.73	-3.18	-3.73	V
J8	Angular accuracy check (cos 270°)	X	X	-2.2	2.2	-2.9	2.9	-2.9	2.9	-3.6	3.6	mV
J9	Angular accuracy check (sin 45° -cos 45° step)	X	X	-3.8	3.8	-5.1	5.1	-5.1	5.1	-6.5	6.5	mV
J10	Angular accuracy check (sin 135° +cos 135° step)	X	X	-3.8	3.8	-5.1	5.1	-5.1	5.1	-6.5	6.5	mV
J11	Angular accuracy check (sin 225° -cos 225° step)	X	X	-3.8	3.8	-5.1	5.1	-5.1	5.1	-6.5	6.5	mV
J12	Angular accuracy check (sin 315° +cos 315° step)	X	X	-3.8	3.8	-5.1	5.1	-5.1	5.1	-6.5	6.5	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
K1	Angular accuracy with input noise check (sin 0°)	XI	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
K2	Angular accuracy with input noise check (cos 0°)	XI	Q	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
K3	Angular accuracy with input noise check (sin 90°)	XI	Q	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
K4	Angular accuracy with input noise check (cos 90°)	XI	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
K5	Angular accuracy with input noise check (sin 180°)	XI	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
K6	Angular accuracy with input noise check (cos 180°)	XI	Q	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V
K7	Angular accuracy with input noise check (sin 270°)	XI	Q	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V
K8	Angular accuracy with input noise check (cos 270°)	XI	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
K9	Angular accuracy with input noise check (sin 45° -cos 45° step)	XI	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
K10	Angular accuracy with input noise check (sin 135° +cos 135° step)	XI	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
K11	Angular accuracy with input noise check (sin 225° -cos 225° step)	XI	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
K12	Angular accuracy with input noise check (sin 315° +cos 315° step)	XI	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L1	Angular accuracy with chassis noise check (sin 0°)	XII	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
L2	Angular accuracy with chassis noise check (cos 0°)	XII	Q	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
L3	Angular accuracy with chassis noise check (sin 90°)	XII	Q	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
L4	Angular accuracy with chassis noise check (cos 90°)	XII	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
L5	Angular accuracy with chassis noise check (sin 180°)	XII	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
L6	Angular accuracy with chassis noise check (cos 180°)	XII	Q	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V
L7	Angular accuracy with chassis noise check (sin 270°)	XII	Q	-3.18	-3.73	-3.18	-3.73	-3.18	-37.3	-3.18	-3.73	V
L8	Angular accuracy with chassis noise check (cos 270°)	XII	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
L9	Angular accuracy with chassis noise check (sin 45° -cos 45° step)	XII	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
L10	Angular accuracy with chassis noise check (sin 135° +cos 135° step)	XII	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
L11	Angular accuracy with chassis noise check (sin 225° -cos 225° step)	XII	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
L12	Angular accuracy with chassis noise check (sin 315° +cos 315° step)	XII	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1	Harmonic rejection (sin 0°)	XIII	Q	-2.2	2.2	-2.9	2.9	-2.9	2.9	-3.6	3.6	mV
M2	Harmonic rejection (cos 0°)	XIII	Q	2.80	3.35	2.80	3.35	2.80	3.35	2.80	3.35	V
M3	Harmonic rejection (sin 90°)	XIII	Q	2.80	3.35	2.80	3.35	2.80	3.35	2.80	3.35	V
M4	Harmonic rejection (cos 90°)	XIII	Q	-2.2	2.2	-2.9	2.9	-2.9	2.9	-3.6	3.6	mV
M5	Harmonic rejection (sin 180°)	XIII	Q	-2.2	2.2	-2.9	2.9	-2.9	2.9	-3.6	3.6	mV
M6	Harmonic rejection (cos 180°)	XIII	Q	-2.80	-3.35	-2.80	-3.35	-2.80	-3.35	-2.80	-3.35	V
M7	Harmonic rejection (sin 270°)	XIII	Q	-2.80	-3.35	-2.80	-3.35	-2.80	-3.35	-2.80	-3.35	V
M8	Harmonic rejection (cos 270°)	XIII	Q	-2.2	2.2	-2.9	2.9	-2.9	2.9	-3.6	3.6	mV
M9	Harmonic rejection (sin 45° -cos 45° step)	XIII	Q	-3.8	3.8	-5.1	5.1	-5.1	5.1	-6.5	6.5	mV
M10	Harmonic rejection (sin 135° +cos 135° step)	XIII	Q	-3.8	3.8	-5.1	5.1	-5.1	5.1	-6.5	6.5	mV
M11	Harmonic rejection (sin 225° -cos 225° step)	XIII	Q	-3.8	3.8	-5.1	5.1	-5.1	5.1	-6.5	6.5	mV
M12	Harmonic rejection (sin 315° +cos 315° step)	XIII	Q	-3.8	3.8	-5.1	5.1	-5.1	5.1	-6.5	6.5	mV
N1	Power supply noise rejection (sin 0°)	XIV	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
N2	Power supply noise rejection (cos 0°)	XIV	Q	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
N3	Power supply noise rejection (sin 90°)	XIV	Q	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
N4	Power supply noise rejection (cos 90°)	XIV	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M5	Power supply noise rejection (sin 180°)	XIV	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
M6	Power supply noise rejection (cos 180°)	XIV	Q	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V
M7	Power supply noise rejection (sin 270°)	XIV	Q	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V
M8	Power supply noise rejection (cos 270°)	XIV	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
M9	Power supply noise rejection (sin 45° -cos 45° step)	XIV	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
M10	Power supply noise rejection (sin 135° +cos 135° step)	XIV	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
M11	Power supply noise rejection (sin 225° -cos 225° step)	XIV	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
M12	Power supply noise rejection (sin 135° +cos 315° step)	XIV	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
M13	Power supply noise rejection (sin 0°)	XIV	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
M14	Power supply noise rejection (cos 0°)	XIV	Q	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
M15	Power supply noise rejection (sin 90°)	XIV	Q	3.18	3.73	3.18	3.73	3.18	3.73	3.18	3.73	V
M16	Power supply noise rejection (cos 90°)	XIV	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
M17	Power supply noise rejection (sin 180°)	XIV	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
M18	Power supply noise rejection (cos 180°)	XIV	Q	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N19	Power supply noise rejection (sin 270°)	XIV	Q	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	-3.18	-3.73	V
N20	Power supply noise rejection (cos 270°)	XIV	Q	-2.3	2.3	-3.0	3.0	-3.0	3.0	-3.6	3.6	mV
N21	Power supply noise rejection (sin 45° -cos 45° step)	XIV	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
N22	Power supply noise rejection (sin 135° +cos 135° step)	XIV	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
N23	Power supply noise rejection (sin 225° -cos 225° step)	XIV	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
N24	Power supply noise rejection (sin 315° +cos 315° step)	XIV	Q	-6.0	6.0	-9.8	9.8	-9.8	9.8	-12.8	12.8	mV
P1	Angular accuracy repeatibility check (sin 0°)	XV	Q	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	mV
P2	Angular accuracy repeatibility check (cos 0°)	XV	Q	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	mV
P3	Angular accuracy repeatibility check (sin 90°)	XV	Q	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	mV
P4	Angular accuracy repeatibility check (cos 90°)	XV	Q	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	mV
P5	Angular accuracy repeatibility check (sin 135° +cos 135° step)	XV	Q	-0.8	0.8	-0.8	0.8	-0.8	0.8	-0.8	0.8	mV
P6	Angular accuracy repeatibility check (sin 315° +cos 315° step)	XV	Q	-0.8	0.8	-0.8	0.8	-0.8	0.8	-0.8	0.8	mV
Q1-Q3	Input current, low level	XVI	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
R1-R3	Input current, high level	XVII	X	0	40	0	40	0	40	0	40	µA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S1-S3	Input clamp diode voltage	XVIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
T1,T2	Analog input impedance check	XIX	Q	0.85	2.40	0.85	2.40	0.85	2.40	0.85	2.40	V_{p-p}
U1-U3	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF
V1-V3	Power supply filter capacitance	XXI	S	0.08	0.22	0.075	0.23	0.075	0.23	0.070	0.24	µF
M1	Power supply current, high level	XXII	S	5	41	5	41	5	41	5	41	mA
M2	Power supply current, low level	XXII	S	5.0	35.5	5.0	35.5	5.0	35.5	5.0	35.5	mA
M3,M4	Power supply current	XXII	S	7.3	12.7	7.0	13.0	7.0	13.0	6.8	13.4	mA
K	Analog output impedance check	XXIII	Q	← See table XXIII for min and max values →								V/V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Test conditions K and J shall also be performed on a 100 pecent "X" test basis at ambient temperatures of 60°C (+5°C, -0°C) and 0°C (-5°C, +0°C). The modules shall meet the initial limits of tables IX and X for the applicable temperature after having temperature stabilized at each temperature for a minimum of 15 minutes.

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4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, F where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions G through J and Q through W which are marked with an "X" or "S" in the test type column shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests C, G through J, and Q through W which are marked with an "X" or "S" in the test type column shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through E and G through X which are marked with an "X" or "S" in the test type column shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	5 V dc ±10% ±15 V dc ±10%
Maximum power dissipation	
5.5 V dc	230 mW
±16.5 V dc	445 mW
Frequency	DC to 400 Hz

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1/ Offset voltage	
Continuous outputs A and B	±5.0 mV
Multiplexed output	
F = 400 Hz	±12.6 mV
1/ Gain	
Continuous outputs A and B	0.987 V dc/V rms +0, -1.0%
Multiplexed output	2.48 V dc/V rms +0, -2.8%

1/ Offset voltage, gain, gain match, noise tolerance, and repeatability are related to the integration period. The values shown are based on the conditions specified.

1/ Gain match	
Continuous outputs A and B	±0.20%
Multiplexed	
F = 400 Hz	±0.16%
1/ Noise tolerance for zero accuracy degradation	
Input noise	0.01 ±0.001 V rms (white noise)
Chassis noise	0.1 ±0.025 V rms (white noise)
Harmonic rejection	5.6 ±0.5 V peak (triangle wave)
Power supply noise	0.01 ±0.001 V rms (white noise)
Slew and settling time	100 µs maximum
Clear time	1.0 ms minimum
1/ Repeatability	2.2 mV maximum
Analog input impedance	5 MΩ minimum
Analog output impedance	2.3 Ω maximum
Logic input current	
Low level	1.73 mA maximum
High level	40 µA maximum

6.4 Module function. This module contains a dual integrator which develops dc output voltages representing the integral of input dc or ac signals.

* 6.4.1 Schematic. A simplified schematic of the module is shown on figure 1. The module accepts either ac or dc signals at both inputs. Three output analog signals are developed. These signals are bipolar dc voltages representing the integrals of the applied signals over an externally controlled acquisition period.

Two of the three outputs are continuously available signals from each of the two integrator channels. The nominal amplitude of each of these signals is determined by the expression:

$$e_o = -.905 \int_{t_1}^{t_2} e_i \, dt$$

The third analog output signal is a multiplexed output of the two integrator channels. The nominal amplitude of each signal at this output is determined by the expression:

$$e_o = G \int_{t_1}^{t_2} e_i \, dt$$

Gain options are available at the multiplexed output. With pin 25 connected to pin 30, the gain factor (G) has the value 2262. With pin 25 connected to pin 21, G has the value 4524.

1/ Offset voltage, gain, gain match, noise tolerance, and repeatability are related to the integration period. The values shown are based on the condition specified.

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The internal logic of the module is compatible with standard TTL logic levels. The control interface consists of three signals, provided by the user, as described below:

a. Integrate hold. This signal causes both integrators to begin integrating their respective incoming signals when it is a logic "H". When the signal is dropped to a logic "L", the integrators are capable of holding the integrated value with a droop not to exceed 0.75 mV/ms (measured at the continuous output) from the initiation of the hold command.

b. Clear. A logic signal of 1.0 ms minimum is required to clear the integrators. A logic "L" is applied to this input line causes the clear function to be activated. A logic "H" deactivates the function.

c. Signal select. A logic "L" applied to this line connects the output channel 1 to the multiplexed output line. A logic "H" connects the output of channel 2 to the switched output line. The settling time required to switch from the output of one channel to the output of the other channel is 100 µs minimum.

The time constants of the two channels are precisely matched to allow processing of 400 Hz resolver signals. A typical timing diagram for processing these signals is shown on figure 2.

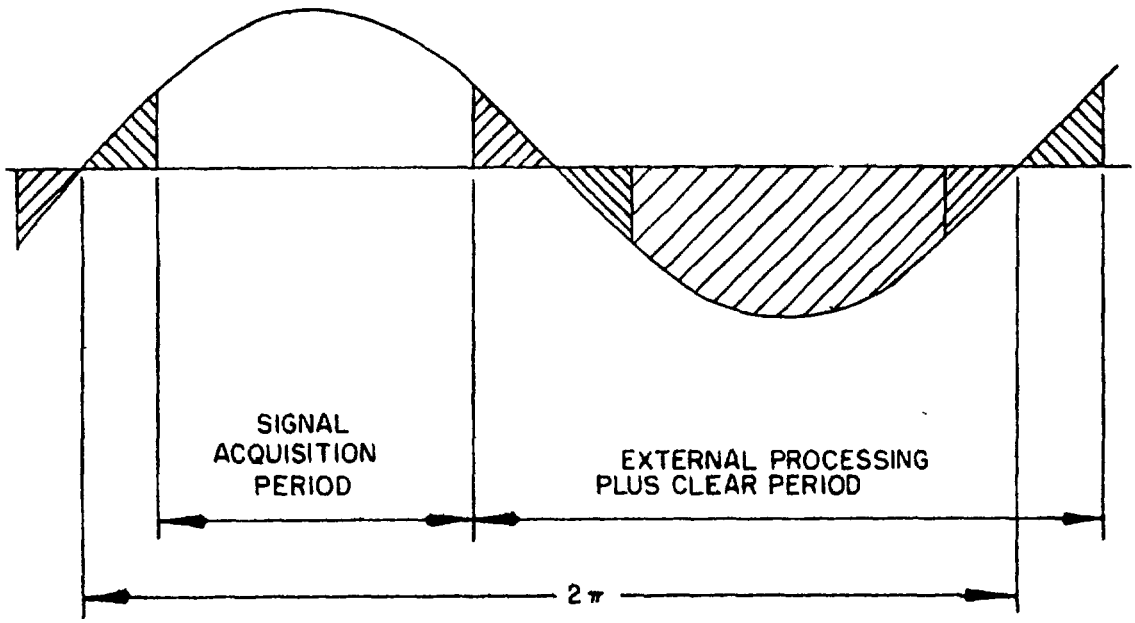


FIGURE 2. Typical timing diagram for processing signals.

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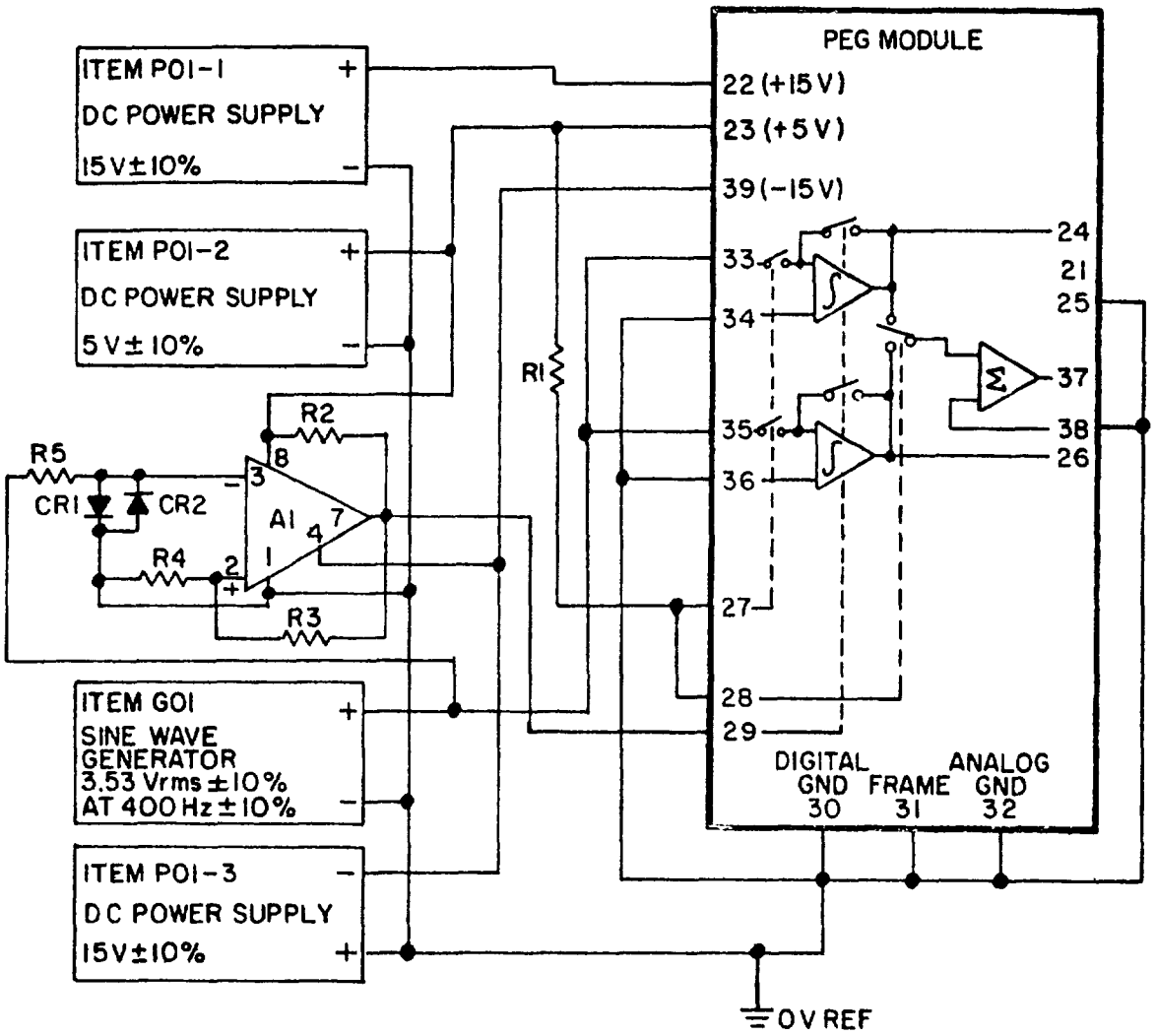
6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES:
1. CR1 = CR2 = 1N4148 or equivalent.
2. R1 = 1 kΩ ±1%, 1/8 W.
3. R2 = 300Ω ±5%, 1/4 W.
4. R3 = 47 kΩ ±1%, 1/8 W.
5. R4 = 470Ω ±1%, 1/8 W.
6. R5 = 10 kΩ ±1%, 1/8 W.
7. A1 = LM111.

FIGURE 3. Operating temperature, component temperature, vibration and life test circuit.

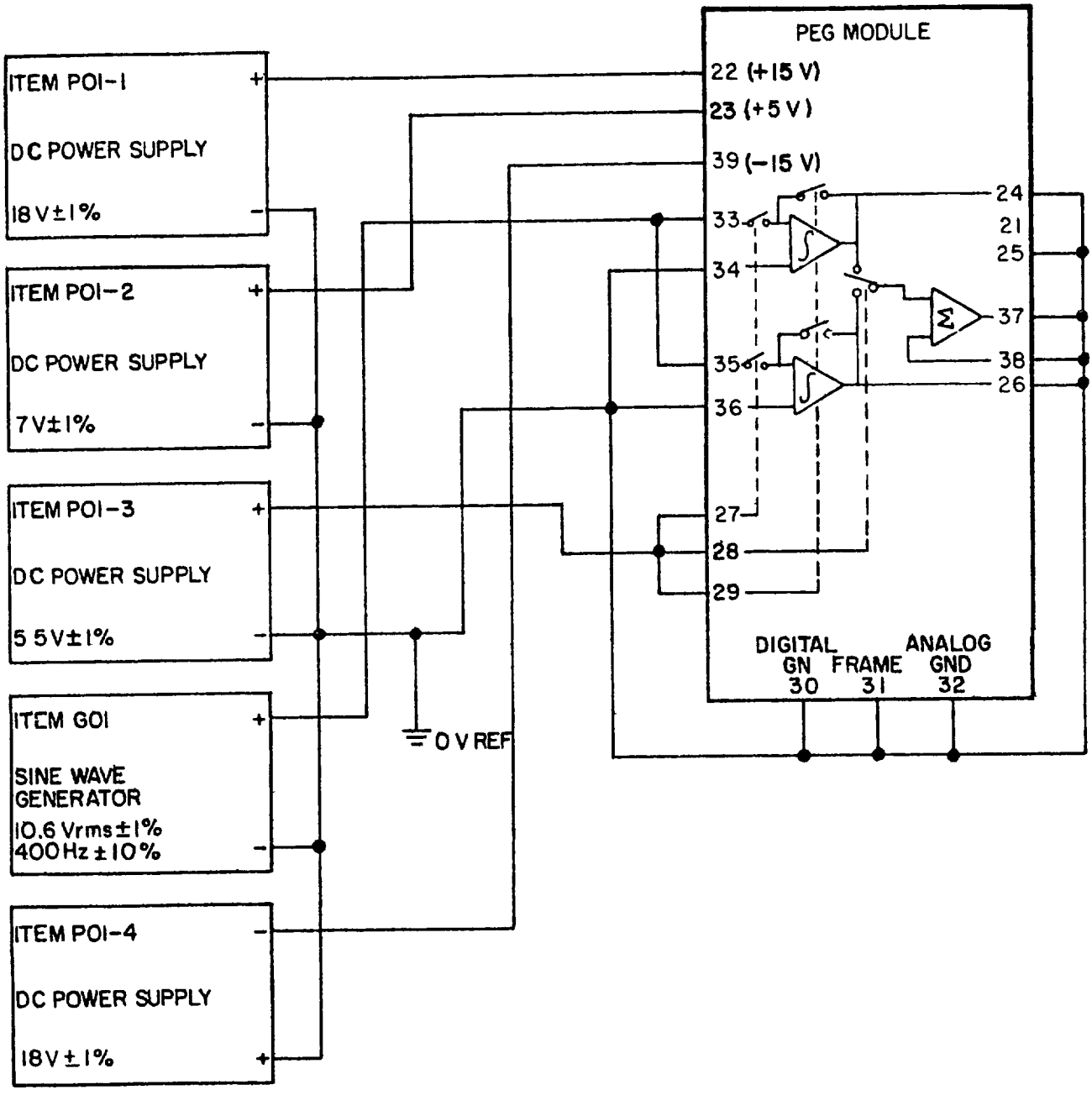
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TABLE II. Maximum stress test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
A	Maximum stress 1/	5	60	s	4

1/ After the maximum stress test, the module shall meet the remaining requirements of table I in the order specified in 4.4.2.

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NOTE: The duration of the maximum stress test shall be as specified in table I.

* FIGURE 4. Maximum stress test circuit for table II.

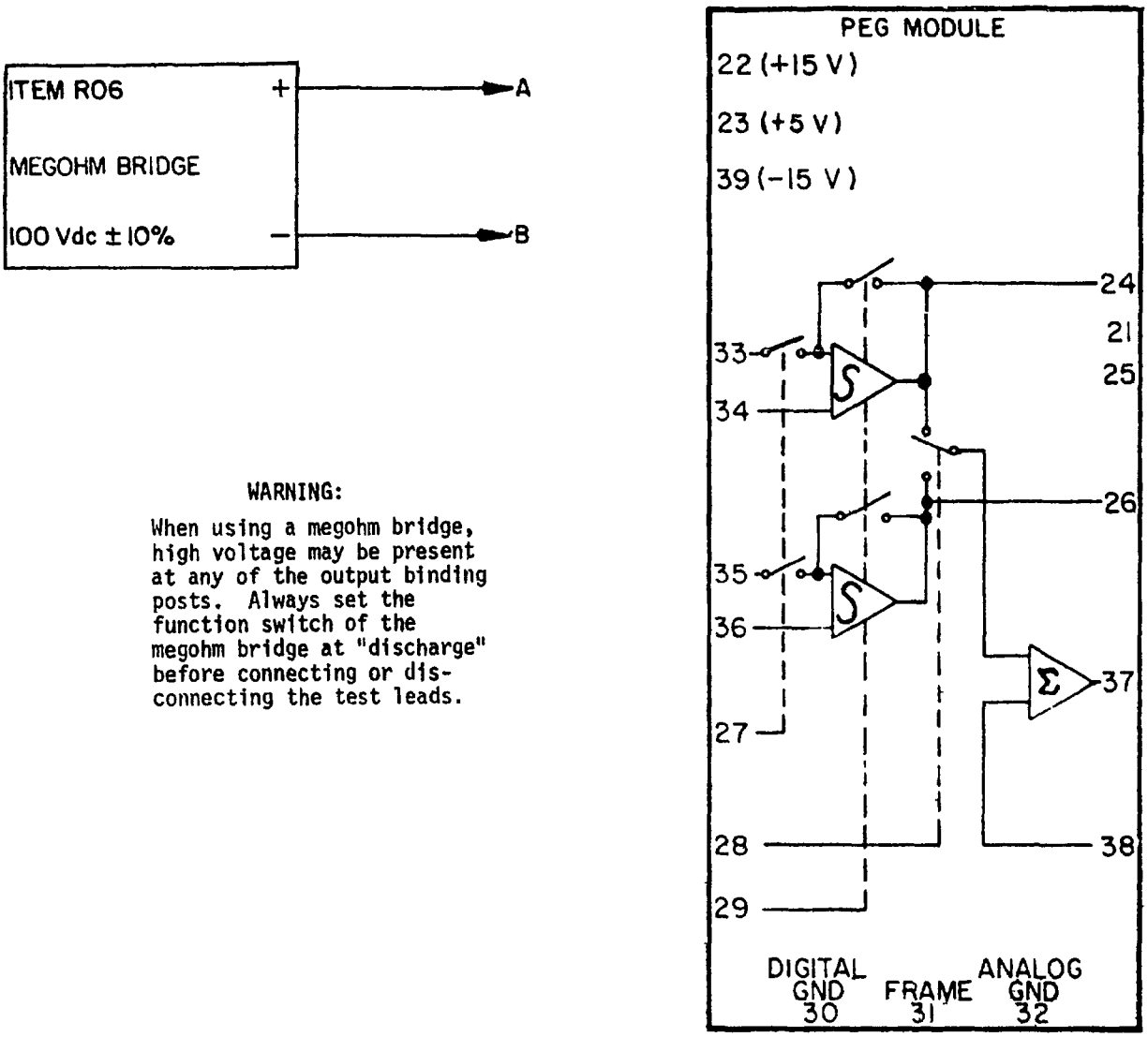
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TABLE III. Isolation test.

Test condition	Parameter	Connect 1/		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation	A31	All other pins	10	∞	MΩ	5
B2	Isolation	A21-A39	All other pins	10	∞	MΩ	5
B3	Isolaton 1/	A1,A3,A5,A7, A9,A11,A13,A15, A17,A19,A40,B21, B23,B25,B27,B29, B31,B33,B35,B37, B39	A2,A4,A6,A8, A10,A12,A14, A16,A18,A20, B22,B24,B26, B28,B30,B32, B34,B36,B38, B40	10	∞	MΩ	5

1/ If either row A1 through A20 or B21 through B40 is not used, the unused pins shall be deleted from test condition B3.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

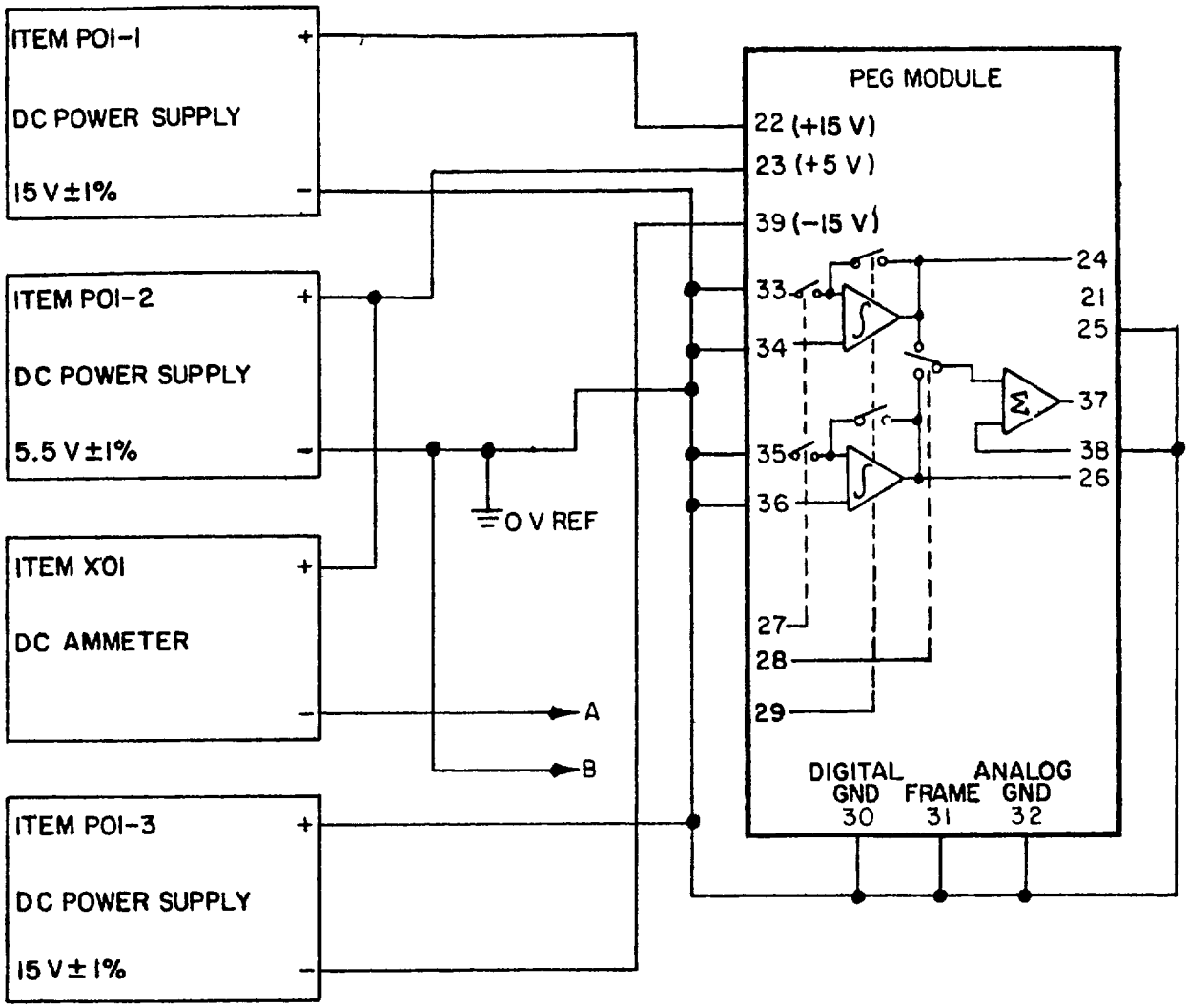
FIGURE 5. Isolation test circuit for table III.

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* TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Input breakdown current	27	28,29	0	1	mA	6
C2	Input breakdown current	28	27,29	0	1	mA	6
C3	Input breakdown current	29	27,28	0	1	mA	6

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

*FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V Ground return isolation test.

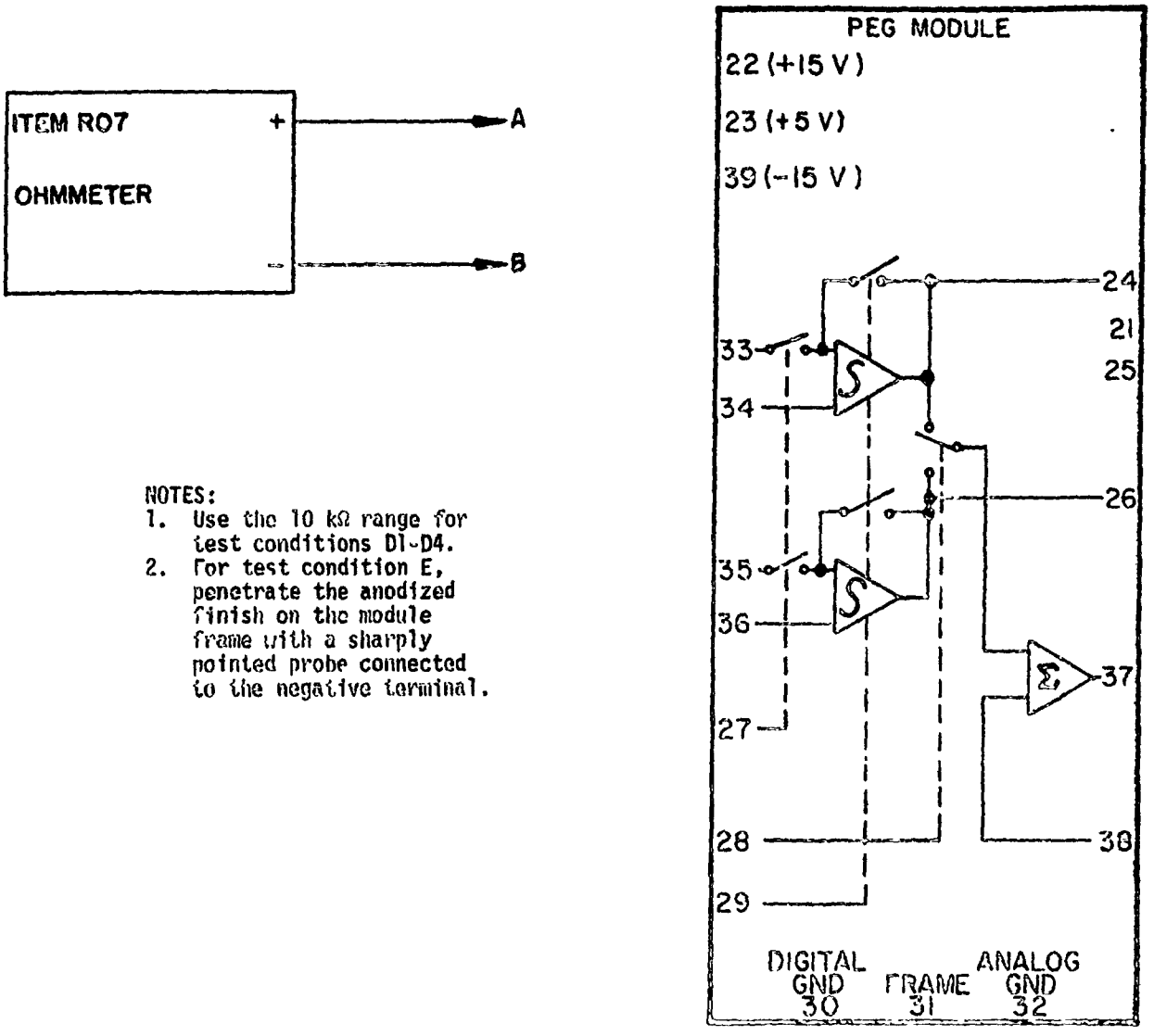
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Ground return isolation	30	32,34,36,38	10	∞	kΩ	7
D2	↓	32	34,36,38	↓	↓	↓	↓
D3		34	36,38				
D4		36	38				

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TABLE VI. Frame continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E	Frame continuity	31	Frame	0	1	Ω	7

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NOTES:
1. Use the 10 kΩ range for test conditions D1-D4.
2. For test condition E, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

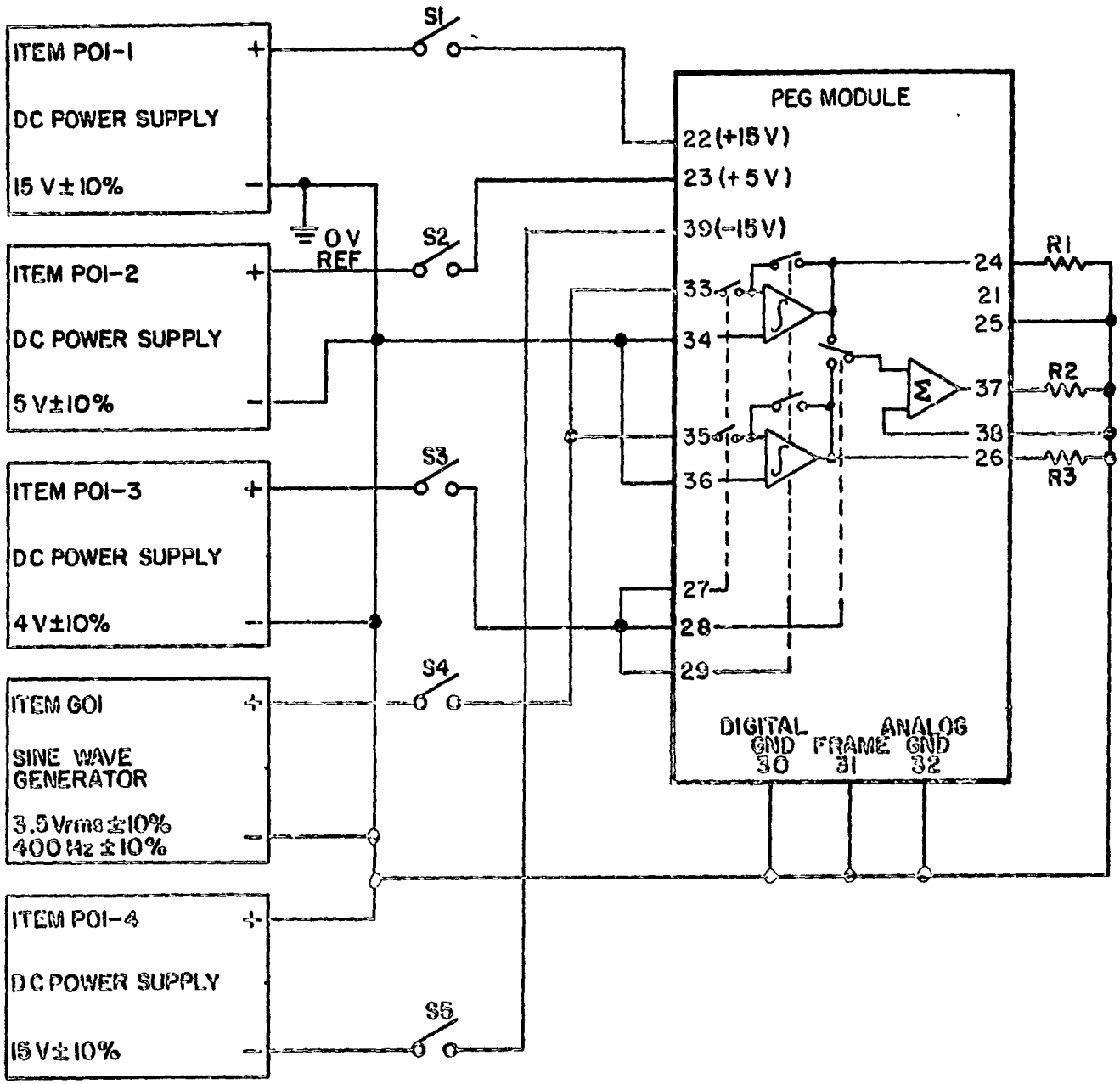
* FIGURE 7. Ground return isolation and frame continuity test circuit for tables V and VI.

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TABLE VII. Voltage sequencing test.

Test condition	Parameter	Switch position					Initial limits 25°±5°C ambient		Unit	Test figure
		S1	S2	S3	S4	S5	Min	Max		
F1	Voltage sequencing	On	Off	Off	Off	Off	5	60	s	8
F2		On	Off	Off	Off	On				
F3		Off	Off	Off	Off	On				
F4		Off	On	Off	Off	Off				
F5		Off	Off	On	Off	Off				
F6		Off	Off	Off	On	Off				
F7		On	Off	Off	On	On				
F8		On	On	Off	On	Off				
F9		Off	On	Off	On	On				
F10		On	Off	On	On	On				

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NOTES:
1. R1 = R2 = R3 = 15 kΩ ±1%, 1/8 W.
2. S1-S5 = SPST.

FIGURE 8. Voltage sequencing test circuit for table VII.

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* TABLE VIII. DC offset voltage and gain check test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	UX to pin	L to pin	M to pin	N to pin	Min	Max		
G1	DC offset voltage at 280 µs	Open or UX	Open	G	37	F	UX	-1.8	1.8	mV	9,10
G2	DC offset voltage at 150 µs	Open or UX	Open	G	↓	↓	↓	-1.8	1.8	mV	↓
G3	DC gain check at 280 µs	UX	G	N	↓	↓	↓	5.60	5.73	V	↓
G4	DC gain check at 150 µs	UX	G	N	↓	↓	↓	-1.5	1.5	mV	↓

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TABLE IX. Angular accuracy check test.

Test condition	Parameter	Connect				Set item F03 to angle (deg)	Initial limits				Unit	Test figure
		H to pin	K to pin	M to pin	N to pin		25°±5°C ambient		0° to 60°C ambient			
							Min	Max	Min	Max		
H1	Angular accuracy check (-sin 0°)	Open	Open	24	UX	0.000	-3.0	3.0	-4.0	4.0	mV	10,11
H2	Angular accuracy check (-cos 0°)			26		0.000	-3.18	-3.73	-3.18	-3.73	V	
H3	Angular accuracy check (-sin 90°)			24		90.000	-3.18	-3.73	-3.18	-3.73	V	
H4	Angular accuracy check (-cos 90°)			26		90.000	-3.0	3.0	-4.0	4.0	mV	
H5	Angular accuracy check (-sin 180°)			24		180.000	-3.0	3.0	-4.0	4.0	mV	
H6	Angular accuracy check (-cos 180°)			26		180.000	3.18	3.73	3.18	3.73	V	
H7	Angular accuracy check (-sin 270°)			24		270.000	3.18	3.73	3.18	3.73	V	
H8	Angular accuracy check (-cos 270°)			26		270.000	-3.0	3.0	-4.0	4.0	mV	
H9	Angular accuracy check (sin 45° -cos 45°)			24	26	45.000	-4.2	4.2	-5.6	-5.6		
H10	Angular accuracy check (sin 135° +cos 135°)	24	26	J	UX	135.000	-2.1	2.1	-2.8	2.8		
H11	Angular accuracy check (sin 225° -cos 225°)	Open	Open	24	26	225.000	-4.2	4.2	-5.6	5.6		
H12	Angular accuracy check (sin 315° +cos 315°)	24	26	J	UX	315.000	-2.1	2.1	-2.8	2.8		

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TABLE X. Angular accuracy check test.

Test condition	Parameter	Connect						Set item F03 to angle (deg)	Initial limits				Unit	Test figure
		A to pin	B to pin	E to pin	L to pin	M to pin	N to pin		25°±5°C ambient		0° to 60°C ambient			
									Min	Max	Min	Max		
J1	Angular accuracy check (sin 0°)	C,UX	D	H	37	F	UX	0.000	-2.2	2.2	-2.9	2.9	mV	10,11
J2	Angular accuracy check (cos 0°)	C,UX	D	↓	↓	↓	↓	0.000	3.18	3.73	3.18	3.73	V	↓
J3	Angular accuracy check (sin 90°)	D,UX	C	↓	↓	↓	↓	90.000	3.18	3.73	3.18	3.73	V	↓
J4	Angular accuracy check (cos 90°)	D,UX	C	↓	↓	↓	↓	90.000	-2.2	2.2	-2.9	2.9	mV	↓
J5	Angular accuracy check (sin 180°)	D	C,UX	↓	↓	↓	↓	180.000	-2.2	2.2	-2.9	2.9	mV	↓
J6	Angular accuracy check (cos 180°)	D	C,UX	↓	↓	↓	↓	180.000	-3.18	-3.73	-3.18	-3.73	V	↓
J7	Angular accuracy check (sin 270°)	C	D,UX	↓	↓	↓	↓	270.000	-3.18	-3.73	-3.18	-3.73	V	↓
J8	Angular accuracy check (cos 270°)	C	D,UX	↓	↓	↓	↓	270.000	-2.2	2.2	-2.9	2.9	mV	↓
J9	Angular accuracy check (sin 45° -cos 45° step)	UX	G	G	↓	↓	↓	45.000	-3.8	3.8	-5.1	5.1	↓	↓
J10	Angular accuracy check (sin 135° +cos 135° step)	D	C	↓	↓	↓	↓	135.000	↓	↓	↓	↓	↓	↓
J11	Angular accuracy check (sin 225° -cos 225° step)	G	UX	↓	↓	↓	↓	225.000	↓	↓	↓	↓	↓	↓
J12	Angular accuracy check (sin 315° +cos 315° step)	C	D	↓	↓	↓	↓	315.000	↓	↓	↓	↓	↓	↓

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TABLE XI. Angular accuracy with input noise check test.

Test condition	Parameter	Connect						Set item F03 to angle (deg)	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	E to pin	L to pin	M to pin	N to pin		Min	Max		
K1	Angular accuracy with input noise check (sin 0°)	C,UX	D	H	37	F	UX	0.000	-2.3	2.3	mV	10,12,13
K2	Angular accuracy with input noise check (cos 0°)	C,UX	D					0.000	3.18	3.73	V	
K3	Angular accuracy with input noise check (sin 90°)	D,UX	C					90.000	3.18	3.73	V	
K4	Angular accuracy with input noise check (cos 90°)	D,UX	C					90.000	-2.3	2.3	mV	
K5	Angular accuracy with input noise check (sin 180°)	D	C,UX					180.000	-2.3	2.3	mV	
K6	Angular accuracy with input noise check (cos 180°)	D	C,UX					180.000	-3.18	-3.73	V	
K7	Angular accuracy with input noise check (sin 270°)	C	D,UX					270.000	-3.18	-3.73	V	
K8	Angular accuracy with input noise check (cos 270°)	C	D,UX					270.000	-2.3	2.3	mV	
K9	Angular accuracy with input noise check (sin 45° -cos 45° step)	UX	G	G				45.000	-6.0	6.0		
K10	Angular accuracy with input noise check (sin 135° -cos 135° step)	D	C					135.000				
K11	Angular accuracy with input noise check (sin 225° -cos 225° step)	G	UX					225.000				
K12	Angular accuracy with input noise check (sin 315° -cos 315° step)	C	D					315.000				

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TABLE XII. Angular accuracy with chassis noise check test.

Test condition	Parameter	Connect						Set item F03 to angle (deg)	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	E to pin	L to pin	M to pin	N to pin		Min	Max		
L1	Angular accuracy with chassis noise check (sin 0°)	C,UX	D	H	37	F	UX	0.000	-2.3	2.3	mV	10,14
L2	Angular accuracy with chassis noise check (cos 0°)	C,UX	D					0.000	3.18	3.73	V	
L3	Angular accuracy with chassis noise check (sin 90°)	D,UX	C					90.000	3.18	3.73	V	
L4	Angular accuracy with chassis noise check (cos 90°)	D,UX	C					90.000	-2.3	2.3	mV	
L5	Angular accuracy with chassis noise check (sin 180°)	D	C,UX					180.000	-2.3	2.3	mV	
L6	Angular accuracy with chassis noise check (cos 180°)	D	C,UX					180.000	-3.18	-3.73	V	
L7	Angular accuracy with chassis noise check (sin 270°)	C	D,UX					270.000	-3.18	-3.73	V	
L8	Angular accuracy with chassis noise check (cos 270°)	C	D,UX					270.000	-2.3	2.3	mV	
L9	Angular accuracy with chassis noise check (sin 45° -cos 45° step)	UX	G	G				45.000	-6.0	6.0		
L10	Angular accuracy with chassis noise check (sin 135° +cos 135° step)	D	C					135.000				
L11	Angular accuracy with chassis noise check (sin 225° -cos 225° step)	G	UX					225.000				
L12	Angular accuracy with chassis noise check (sin 315° +cos 315° step)	C	D					315.000				

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TABLE XIII. Harmonic rejection test.

Test condition	Parameter	Connect						Set item F03 to angle (deg)	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	E to pin	L to pin	M to pin	N to pin		Min	Max		
M1	Harmonic rejection (sin 0°)	C,UX	D	H	37	F	UX	0.000	-2.2	2.2	mV	10,15
M2	Harmonic rejection (cos 0°)	C,UX	D					0.000	2.80	3.35	V	
M3	Harmonic rejection (sin 90°)	D,UX	C					90.000	2.80	3.35	V	
M4	Harmonic rejection (cos 90°)	D,UX	C					90.000	-2.2	2.2	mV	
M5	Harmonic rejection (sin 180°)	D	C,UX					180.000	-2.2	2.2	mV	
M6	Harmonic rejection (cos 180°)	D	C,UX					180.000	-2.80	-3.35	V	
M7	Harmonic rejection (sin 270°)	C	D,UX					270.000	-2.80	-3.35	V	
M8	Harmonic rejection (cos 270°)	C	D,UX					270.000	-2.2	2.2	mV	
M9	Harmonic rejection (sin 45° -cos 45° step)	UX	G	G				45.000	-3.8	3.8		
M10	Harmonic rejection (sin 135° -cos 135° step)	D	C					135.00				
M11	Harmonic rejection (sin 225° -cos 225° step)	G	UX					225.000				
M12	Harmonic rejection (sin 315° -cos 315° step)	C	D					315.000				

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TABLE XIV. Power supply noise rejection test.

Test condition	Parameter	Connect						Set item F03 to angle (deg)	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	E to pin	L to pin	M to pin	N to pin		Min	Max		
N1	Power supply noise rejection (sin 0°)	C,UX	D	H	37	F	UX	0.000	-2.3	2.3	mV	10,13,16
N2	Power supply noise rejection (cos 0°)	C,UX	D					0.000	3.18	3.73	V	
N3	Power supply noise rejection (sin 90°)	D,UX	C					90.000	3.18	3.73	V	
N4	Power supply noise rejection (cos 90°)	D,UX	C					90.000	-2.3	2.3	mV	
N5	Power supply noise rejection (sin 180°)	D	C,UX					180.000	-2.3	2.3	mV	
N6	Power supply noise rejection (cos 180°)	D	C,UX					180.000	-3.18	-3.73	V	
N7	Power supply noise rejection (sin 270°)	C	D,UX					270.000	-3.18	-3.73	V	
N8	Power supply noise rejection (cos 270°)	C	D,UX					270.000	-2.3	2.3	mV	
N9	Power supply noise rejection (sin 45° -cos 45° step)	UX	G					45.000	-6.0	6.0		
N10	Power supply noise rejection (sin 135° +cos 135° step)	D	C					135.000				
N11	Power supply noise rejection (sin 225° -cos 225° step)	G	UX					225.000				
N12	Power supply noise rejection (sin 315° +cos 315° step)	C	D					315.000				
N13	Power supply noise rejection (sin 0°)	C,UX	D	H				0.000	-2.3	2.3		
N14	Power supply noise rejection (cos 0°)	C,UX	D					0.000	3.18	3.73	V	
N15	Power supply noise rejection (sin 90°)	D,UX	C					90.000	3.18	3.73	V	
N16	Power supply noise rejection (cos 90°)	D,UX	C					90.000	-2.3	2.3	mV	

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TABLE XIV. Power supply noise rejection test - Continued.

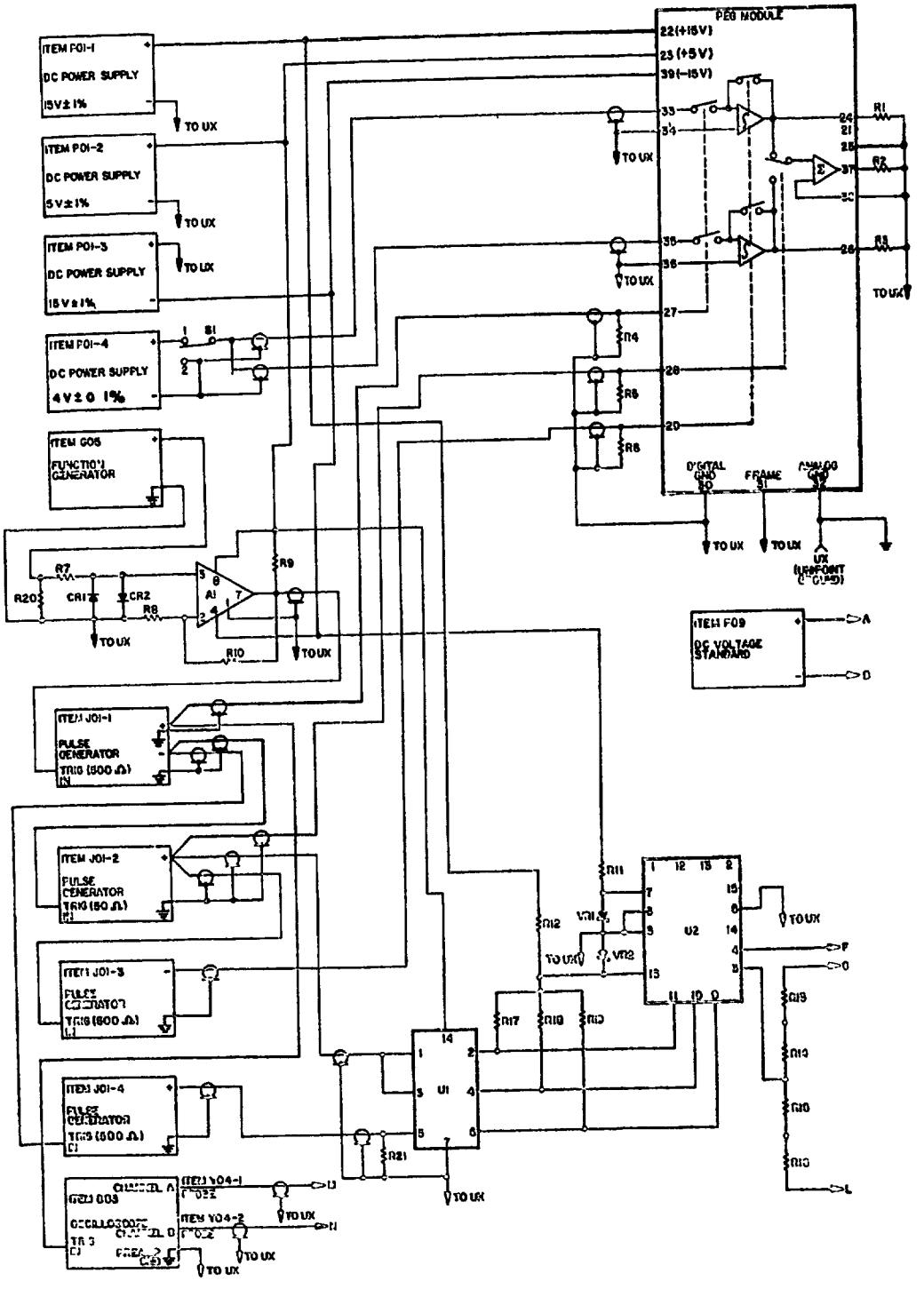
Test condition	Parameter	Connect						Set item F03 to angle (deg)	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	E to pin	L to pin	M to pin	N to pin		Min	Max		
N17	Power supply noise rejection (sin 180°)	D	C,UX	H	37	F	UX	180.000	-2.3	2.3	mV	10,13,16
N18	Power supply noise rejection (cos 180°)	D	C,UX	↓	↓	↓	↓	180.000	-3.18	-3.73	V	↓
N19	Power supply noise rejection (sin 270°)	C	D,UX	↓	↓	↓	↓	270.000	-3.18	-3.73	V	↓
N20	Power supply noise rejection (cos 270°)	C	D,UX	↓	↓	↓	↓	270.000	-2.3	2.3	mV	↓
N21	Power supply noise rejection (sin 45° -cos 45° step)	UX	G	G	↓	↓	↓	45.000	-6.0	6.0	↓	↓
N22	Power supply noise rejection (sin 135° +cos 135° step)	D	C	↓	↓	↓	↓	135.000	↓	↓	↓	↓
N23	Power supply noise rejection (sin 225° -cos 225° step)	G	UX	↓	↓	↓	↓	225.000	↓	↓	↓	↓
N24	Power supply noise rejection (sin 315° +cos 315° step)	C	D	↓	↓	↓	↓	315.000	↓	↓	↓	↓

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TABLE XV. Angular accuracy repeatibility check test.

Test condition	Parameter	Connect						Set item F03 to angle (deg)	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	E to pin	L to pin	M to pin	N to pin		Min	Max		
P1	Angular accuracy repeatibility check (sin 0°)	C,D,UX	Open	H	37	F	UX	0.000	-0.5	0.5	mV	10,17
P2	Angular accuracy repeatibility check (cos 0°)	C,UX	D					0.000				
P3	Angular accuracy repeatibility check (sin 90°)	D,UX	C					90.000				
P4	Angular accuracy repeatibility check (cos 90°)	C,D,UX	Open					90.000				
P5	Angular accuracy repeatibility check (sin 135° +cos 135° step)	D	C	G				135.000	-0.8	0.8		
P6	Angular accuracy repeatibility check (sin 315° +cos 315° step)	C	D	G				315.000	-0.8	0.8		

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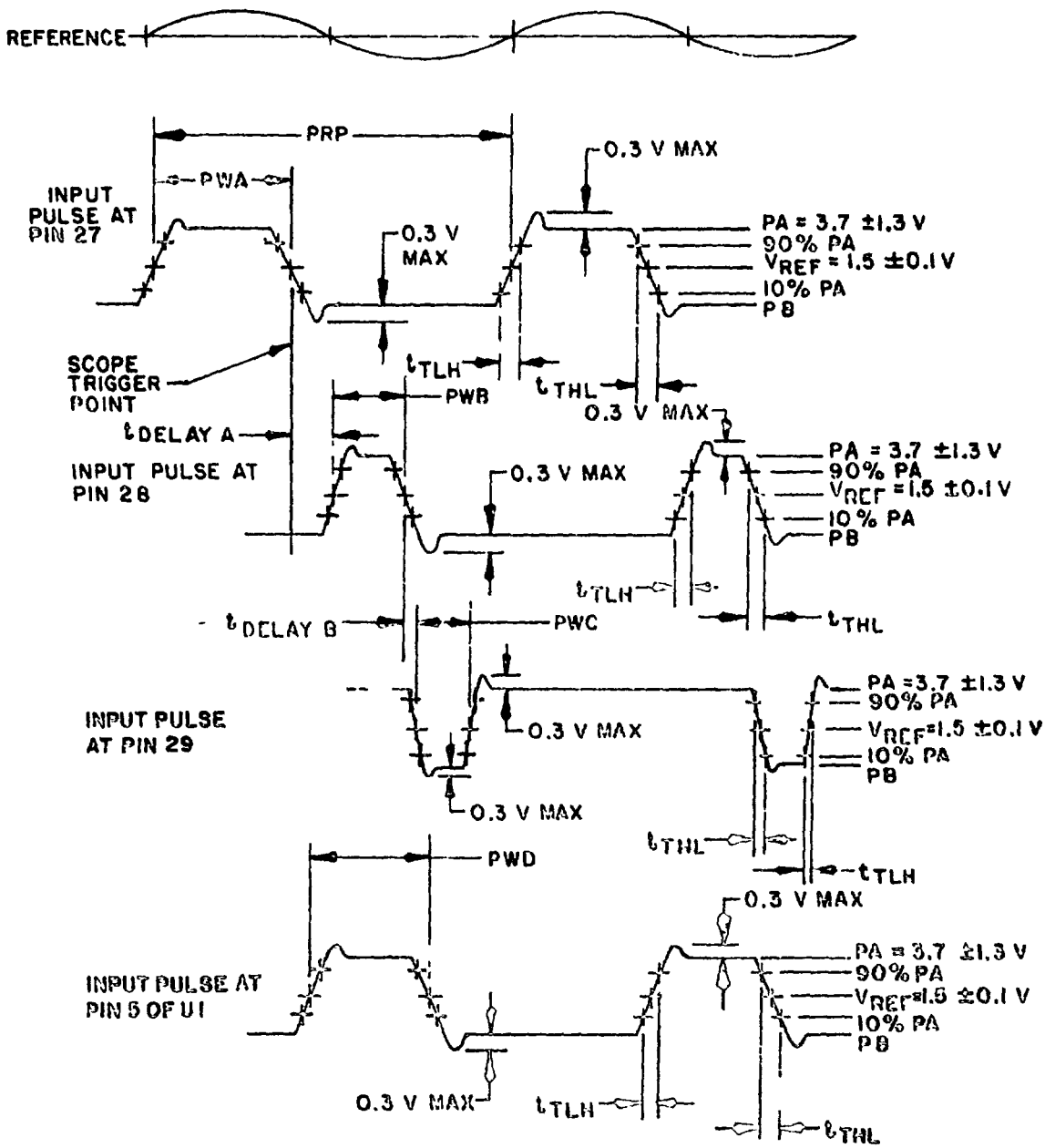
☆ FIGURE 9. DC offset voltage and gain check test circuit for table VIII.

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NOTES:
1. ⏚ TO UX = Indicates individual connections that must be made at the unipoint ground (pin 32).
2. All equipment except item D03 must be isolated from earth ground.
3. A1 = LM111.
4. CR1 = CR2 = 1N4148 or equivalent.
5. R1-R3 = 15 kΩ ±1%, 1/8 W.
6. R4-R6, R20, R21 = 51Ω ±5%, 1/2 W.
7. R7 = 10 kΩ ±1%, 1/8 W.
8. R8 = 470Ω ±1%, 1/8 W.
9. R9 = 300Ω ±5%, 1/4 W.
10. R10 = 47 kΩ ±1%, 1/8 W.
11. R11 = R12 = 750Ω ±5%, 1 W.
12. R13 = 12.5 kΩ ±0.1%, 1/10 W.
13. R14 = R15 = 30 kΩ ±0.05% matched to ±0.02%, 1/10 W.
14. R16 = 45 kΩ ±0.1%, 1/10 W.
15. R17-R19 = 20 kΩ ±5%, 1/8 W.
16. S1 = SPDT switch.
17. VR1 = VR2 = 1N4461 or equivalent.
18. U1 = 5417 HEX open-collector buffer.
19. U2 = CD4053B analog switch.
20. Test conditions G1-G4:
    a. Adjust item G05 for a 5.0 ±0.5 V_{p-p}, 400 ±1 Hz square wave.
    b. Apply the input pulses as specified on figure 10.
    c. Set item D03 trigger to EXT(-)DC, sweep to 50 µs/cm, and input select to (DC) for both channels.
    d. Set switch S1 to position 2.
    e. Set item D03 vertical sensitivity to 2 mV/cm.
    f. Adjust the vertical position for a 0 V reading 380 µs from the start of the trace.
    g. Measure and record the vertical deflection of item D03 trace 150 µs from the start of the trace.
    h. The measured value shall satisfy the limits of test condition G1.
    i. Measure and record the vertical deflection of item D03 trace 280 µs from the start of the trace.
    j. The measured value shall satisfy the limits of test condition G2.
    k. Subtract the value recorded in 20i from the value recorded in 20g. Record the difference and observe polarity.
    l. Set switch S1 to position 1.
    m. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 280 µs from the start of the trace.
    n. The output voltage of item P09 shall satisfy the limits of test condition G3.
    o. Measure and record the vertical deflection of item D03, 150 µs from start of the trace.
    p. Measure and record the vertical deflection of item D03, 280 µs from start of the trace.
    q. Subtract the value recorded in 20p from the value recorded in 20o. Record the difference and observe polarity.
    r. Subtract the recorded value in 20q from the recorded value in 20k.
    s. The difference shall satisfy the limits of test condition G4.
21. The vertical amplifier used in item D03 shall be a differential amplifier.

* FIGURE 9. DC offset voltage and gain check test circuit for table VIII - Continued.

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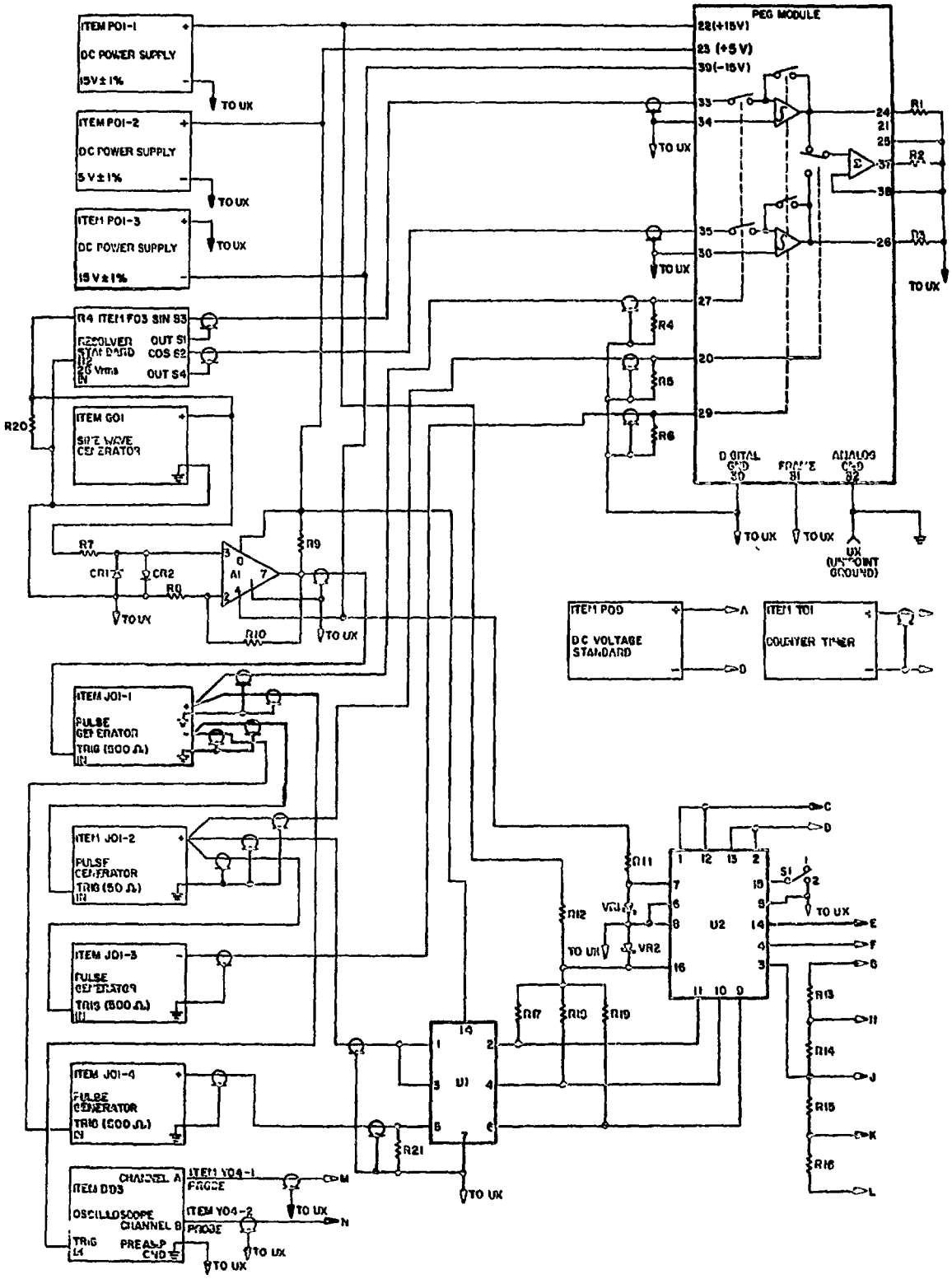
☆ FIGURE 10. DC offset voltage and gain check, angular accuracy check, angular accuracy with input noise check, angular accuracy with chassis noise check, harmonic rejection, power supply noise rejection, and angular accuracy repeatibility check test circuit waveforms for tables VIII through XV.

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NOTES:
1. Input pulse parameters:
   PRP is determined by reference signal.
   PWA = 1110 ±5 µs.
   PWB = 120 ±5 µs.
   PWC = 72 $^{+0}_{-4}$ µs.
   PWD = 320 ±10 µs.
   Pulse base (PB) = 0 ±0.4 V.
   $t_{DELAY\ A}$ = 180 ±5 µs.
   $t_{DELAY\ B}$ = 40 ±5 µs.
   $t_{TLH} = t_{THL}$ = 50 ns (maximum).
2. Adjust item J01-1 negative output voltage for PA = 3.7 ±1.3 V and PB = 0.4 ±0.4 V.

* FIGURE 10. DC offset voltage and gain check, angular accuracy check, angular accuracy with input noise check, angular accuracy with chassis noise check, harmonic rejection, power supply noise rejection, and angular accuracy repeatibility check test circuit waveforms for tables VIII through XV - Continued.

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★ FIGURE 11. Angular accuracy check test circuit for tables IX and X.

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NOTES:
1. ⏚ TO UX = Indicates individual connections that must be made at the unipoint ground (pin 32).
2. All equipment except item D03 must be isolated from earth ground.
3. A1 = LM111.
4. CR1 = CR2 = 1N4148, or equivalent.
5. R1-R3 = 15 kΩ ±1%, 1/8 W.
6. R4-R6 = R20 = R21 = 51Ω ±5%, 1/2 W.
7. R7 = 10 kΩ ±1%, 1/8 W.
8. R8 = 470Ω ±1%, 1/8 W.
9. R9 = 300Ω ±5%, 1/4 W.
10. R10 = 47 kΩ ±1%, 1/8 W.
11. R11 = R12 = 750Ω ±5%, 1 W.
12. R13 = 12.5 kΩ ±0.1%, 1/10 W.
13. R14 = R15 = 30 kΩ ±0.05% matched to ±0.02%, 1/10 W.
14. R16 = 45 kΩ ±0.1%, 1/10 W.
15. R17-R19 = 20 kΩ ±5%, 1/8 W.
16. S1 = SPDT switch.
17. VR1 = VR2 = 1N4461 or equivalent.
18. U1 = 5417 HEX open-collector buffer.
19. U2 = CD4053B analog switch.
20. Test conditions H1, H4, H5, H8-H12:
    a. Set the item F03 "TYPE" switch to "RESOLVER". Set the E_S(MAX) "OUTPUT" switch to 26 V and the decade "DEGREES" switches to 0.0000°.
    b. Adjust item G01 for 3.535 ±0.035 Vrms, 400 ±1 Hz at the cosine output terminals of item F03.
    c. Apply the input pulses as specified on figure 10.
    d. Set item D03 trigger to EXT(-)DC, sweep to 50 µs/cm, and input select to (DC) for both channels.
    e. Set switch S1 to position 1.
    f. Connect the equipment as specified in table IX for the appropriate test condition.
    g. Set item D03 vertical sensitivity to 1 mV/cm.
    h. Adjust the vertical position for a 0 V reading, 380 µs from the start of the trace.
    i. Measure the vertical deflection of item D03 trace, 150 µs from the start of the trace.
    j. The measured value shall satisfy the limits of table IX.
21. Test conditions H2, H3, H6, and H7:
    a. Perform test setup per notes 20a - 20f.
    b. Set the item D03 vertical sensitivity to 1 V/cm.
    c. Perform test per notes 20h and 20j.
22. Test conditions J1, J2, J5, and J6:
    a. Perform test setup per notes 20a - 20e.
    b. Connect the equipment as specified in table X for the appropriate test condition.
    c. Set the item D03 vertical sensitivity to 1 mV/cm. Adjust the vertical position for a 0 V reading, 380 µs from the start of the trace.
    d. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 280 µs from the start of the trace.
    e. The magnitude of the output voltage of item P09 shall satisfy the limits of table X for test conditions J2 and J6(disregard the polarities of the limits).
    f. Concurrently measure the vertical deflection of item D03 trace, 150 µs from the start of the trace for test conditions J1 and J5.
    g. The measured value shall satisfy the limits of table X.

* FIGURE 11. Angular accuracy check test circuit for tables IX and X - Continued.

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23. Test conditions J3, J4, J7, and J8:

    a. Perform test setup per notes 20a - 20e.
    b. Connect the equipment as specified in table X for the appropriate test condition.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 V reading, 380 µs from the start of the trace.
    d. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 150 µs from the start of the trace for test conditions J3 and J7.
    e. The magnitude of the output voltage of item P09 shall satisfy the limits of table X for test conditions J3 and J7 (disregard the polarities of the limits).
    f. Concurrently measure the vertical deflection of item D03 trace 280 µs from the start of the trace for test conditions J4 and J8.
    g. The measured value shall satisfy the limits of table X.

24. Test conditions J9 and J11:

    a. Perform test setup per notes 20a - 20e.
    b. Connect the equipment as specified in table X for appropriate test condition.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 V reading, 380 µs from the start of the trace.
    d. Adjust the output voltage of item P09 for a 0 ±0.5 mV indication on item D03, 150 µs from the start of the trace.
    e. The step between 150 µs and 280 µs from the beginning of the trace shall satisfy the limits of table X.

25. Test conditions J10 and J12:

    a. Perform test setup per notes 20a - 20d.
    b. Connect the equipment as specified in table X for appropriate test condition.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 V reading, 380 µs from the start of the trace.
    d. Set switch S1 to position 2.
    e. Adjust the output voltage of item P09 so that the step between 150 µs and 280 µs from the beginning of the trace is 0 ±0.5 mV.
    f. Simultaneously, measure the average value of the vertical deflection, 280 µs from the start of the trace.
    g. The value measured in 25f shall satisfy the limits of table X.

26. The vertical amplifier used in item D03 shall be a differential amplifier.

* FIGURE 11. Angular accuracy check test circuit for tables IX and X - Continued.

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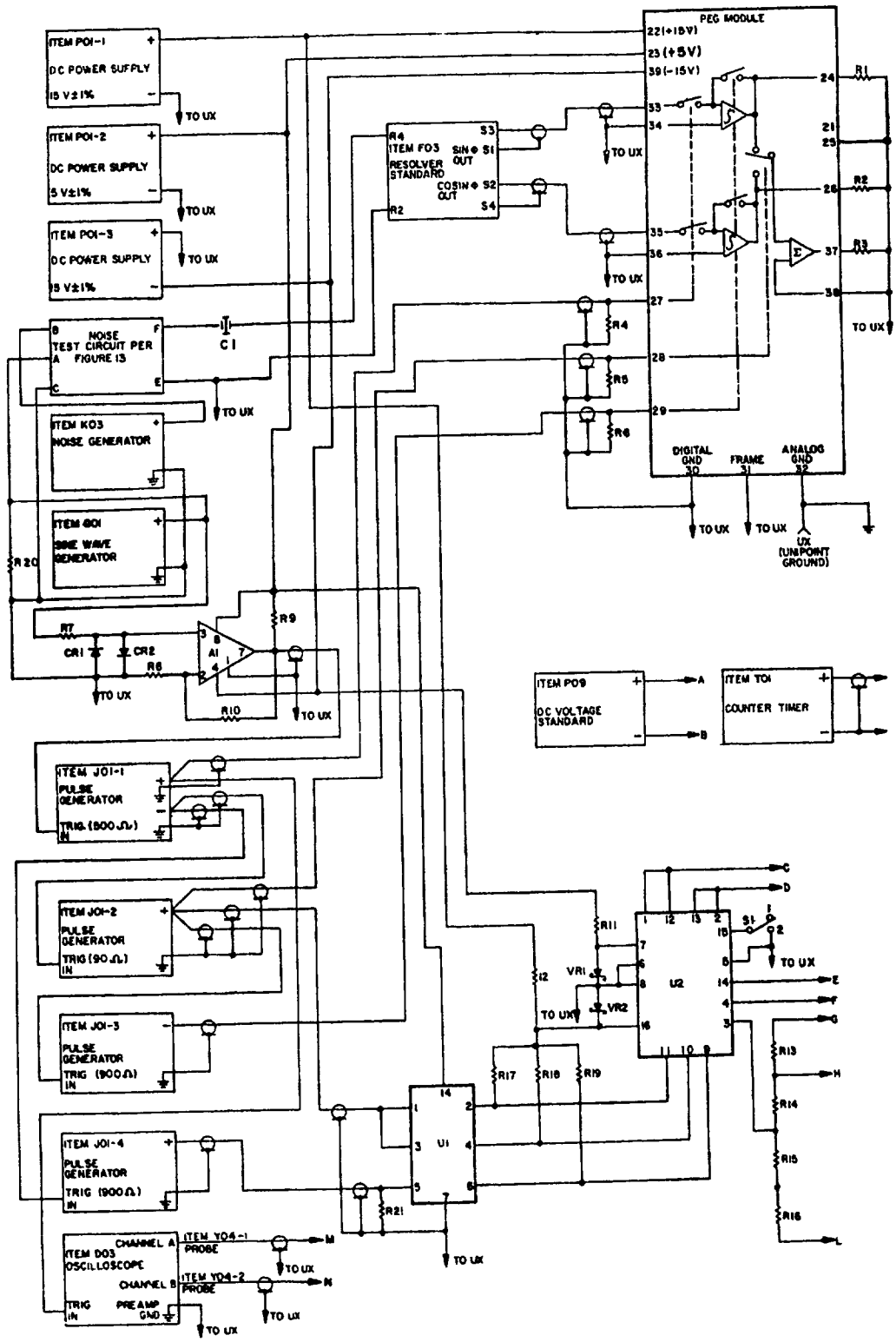


FIGURE 12. Angular accuracy with input noise check test circuit for table XI.

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NOTES:
1. ↓ TO UX = Indicates individual connections that must be made at the unipoint ground (pin 32).
2. All equipment except item D03 must be isolated from earth ground.
3. A1 = LM111.
4. CR1 = CR2 = 1N4148, or equivalent.
5. R1-R3 = 15 kΩ ±1%, 1/8 W.
6. R4-R6 = R20 = R21 = 51Ω ±5%, 1/2 W.
7. R7 = 10 kΩ ±1%, 1/8 W.
8. R8 = 470Ω ±1%, 1/8 W.
9. R9 = 300Ω ±5%, 1/4 W.
10. R10 = 47 kΩ ±1%, 1/8 W.
11. R11 = R12 = 750Ω ±5%, 1 W.
12. R13 = 12.5 kΩ ±0.1%, 1/10 W.
13. R14 = R15 = 30 kΩ ±0.05% matched to ±0.02%, 1/10 W.
14. R16 = 45 kΩ ±0.1%, 1/10 W.
15. R17-R19 = 20 kΩ ±5%, 1/8 W.
16. C1 = 680 µF $^{+75}_{-10}$%, 5 Vdc N.P.
17. S1 = SPDT switch.
18. VR1 = VR2 = 1N4461, or equivalent.
19. U1 = 5417 HEX open-collector buffer.
20. U2 = CD4053B analog switch.
21. Test conditions K1, K2, K5, and K6:

    a. Set the item F03 "TYPE" switch to "RESOLVER". Set the E_S(MAX) "OUTPUT" switch to 26 V and the decade "DEGREES" switches to 0.0000°.
    b. Adjust item G01 for 3.535 ±0.035 Vrms, 400 ±1 Hz at the cosine output terminals of item F03.
    c. Adjust item K03 for 0.01 ±0.001 Vrms "WHITE NOISE" at the cosine output terminals of item F03. Set item K03 band switch to 5 Hz to 20 kHz.
    d. Apply the input pulses as specified on figure 10.
    e. Set item D03 trigger to EXT(-)DC, sweep to 50 µs/cm, and input select to (DC) for both channels.
    f. Set switch S1 to position 1.
    g. Connect the equipment as specified in table XI for the appropriate test condition.
    h. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 V reading, 380 µs from the start of the trace.
    i. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 280 µs from the start of the trace.
    j. The magnitude of the output voltage of item P09 shall satisfy the limits of test conditions K2 and K6(disregard the polarities of the limits).
    k. Concurrently measure the vertical deflection of item D03 trace, 150 µs from the start of the trace for test conditions K1 and K5, respectively.
    l. The measured value shall satisfy the limits of the test conditions.

FIGURE 12. Angular accuracy with input noise check test circuit for table XI - Continued.

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22. Test conditions K3, K4, K7, and K8:

    a. Perform test setup per notes 21a - 21f.
    b. Connect the equipment as specified in table XI for the appropriate test conditions.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 V reading, 380 µs from the start of the trace.
    d. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 150 µs from the start of the trace for K3 and K7.
    e. The magnitude of the output voltage of item P09 shall satisfy the limits of test conditions K3 and K7 (disregard the polarities of the limits).
    f. Concurrently measure the vertical deflection of item D03 trace, 280 µs from the start of the trace for test conditions K4 and K8, respectively.
    g. The measured value shall satisfy the limits of the test conditions.

23. Test conditions K9 and K11:

    a. Perform test setup per notes 21a - 21f.
    b. Connect the equipment as specified in table XI for the appropriate test condition.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 V reading, 380 µs from the start of the trace.
    d. Adjust the output voltage of item P09 for a 0 ±0.5 mV indication on item D03, 150 µs from the start of the trace.
    e. The step between 150 µs and 280 µs from the beginning of the trace shall satisfy the limits of the test conditions.

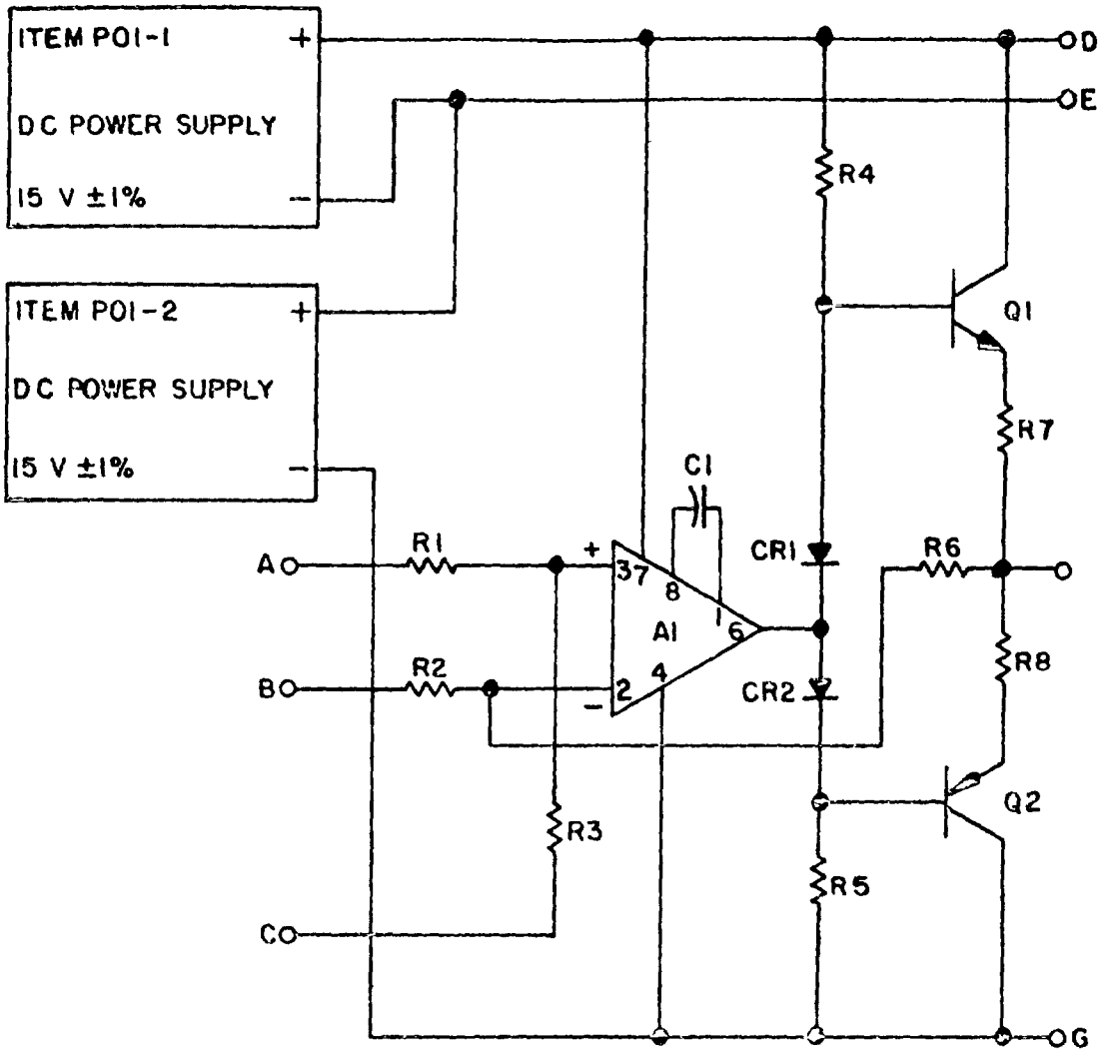
24. Test conditions K10 and K12:

    a. Perform test setup per notes 21a - 21e.
    b. Connect the equipment as specified in table XI for appropriate test condition.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 V reading, 380 µs from the start of the trace.
    d. Set switch S1 to position 2.
    e. Adjust the output voltage of item P09 so that the step between 150 µs and 280 µs from the beginning of the trace is 0 ±0.5 mV.
    f. Simultaneously, measure the average value of the vertical deflection of item D03, 280 µs from the start of the trace.
    g. The value measured in 24f shall satisfy the limits of the test conditions.

25. The vertical amplifier used in item D03 shall be a differential amplifier.

FIGURE 12. Angular accuracy with input noise check test circuit for table XI - Continued.

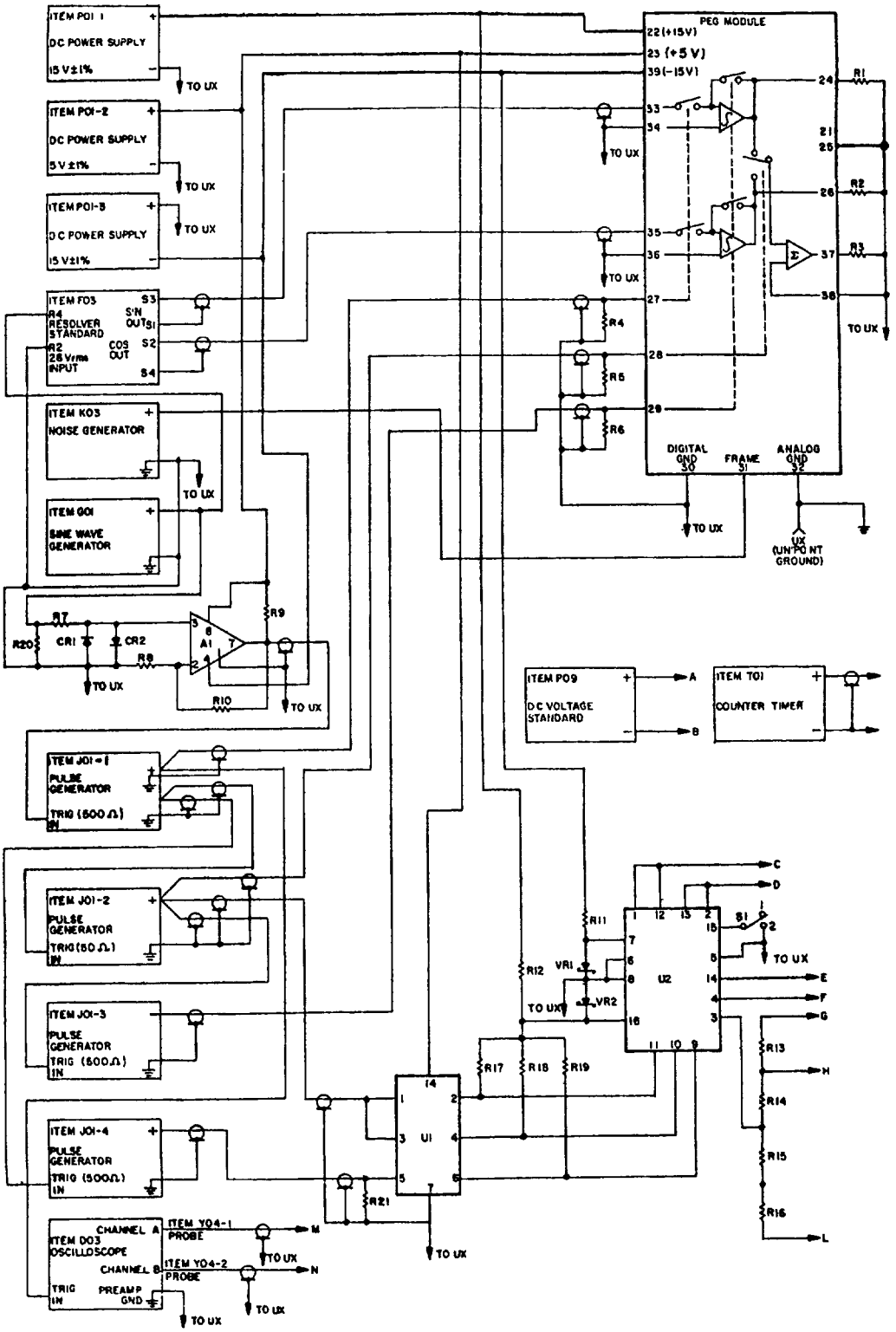
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NOTES:
1. R1-R3, R6 = 51 kΩ ±1%, 1/8 W.
2. R4 = R5 = 10 kΩ ±5%, 1/4 W.
3. R7 = R8 = 10Ω ±5%, 1/2 W.
4. Q1 = 2N5150, or equivalent.
5. Q2 = 2N5149, or equivalent.
6. C1 = 30 pF ±10%, 50 V.
7. A1 = LM108A.
8. CR1 = CR2 = 1N4148, or equivalent.

FIGURE 13. Noise test circuit for figures 12 and 16.

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* FIGURE 14. Angular accuracy with chassis noise check test circuit for table XII.

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NOTES:
1. ↓ TO UX = Indicates individual connections that must be made at the unipoint ground (pin 32).
2. All equipment except item D03 must be isolated from earth ground.
3. A1 = LM111.
4. CR1 = CR2 = 1N4148, or equivalent.
5. R1 = R2 = R3 = 15 kΩ ±1%, 1/8 W.
6. R4 = R5 = R6 = R20 = R21 = 51Ω ±5%, 1/2 W.
7. R7 = 10 kΩ ±1%, 1/8 W.
8. R8 = 470Ω ±1%, 1/8 W.
9. R9 = 300Ω ±5%, 1/4 W.
10. R10 = 47 kΩ ±1%, 1/8 W.
11. R11 = R12 = 750Ω ±5%, 1 W.
12. R13 = 12.5 kΩ ±0.1%, 1/10 W.
13. R14 = R15 = 30 kΩ ±0.05% matched to ±0.02%, 1/10 W.
14. R16 = 45 kΩ ±0.1%, 1/10 W.
15. R17 = R18 = R19 = 20Ω ±5%, 1/8 W.
16. S1 = SPDT switch.
17. VR1 = VR2 = 1N4461, or equivalent.
18. U1 = 5417 HEX open-collector buffer.
19. U2 = CD4053B analog switch.
20. Test conditions L1, L2, L5, and L6:
    a. Set item F03 "TYPE" switch to "RESOLVER". Set the E_S(MAX) "OUTPUT" switch to 26 V and the decade "DEGREES" switches to 0.0000°.
    b. Adjust item G01 for 3.535 ±0.035 Vrms, 400 ±1 Hz at the cosine output terminals of item F03.
    c. Adjust item K03 for 0.1 ±0.025 Vrms "WHITE NOISE" between pin 31 and UX. Set item K03 bandwidth to the 5 Hz to 20 kHz.
    d. Apply the input pulses as specified on figure 10.
    e. Set item D03 trigger to EXT(-)DC, sweep to 50 µs/cm, and input select to (DC) for both channels.
    f. Set switch S1 to position 1.
    g. Connect the equipment as specified in table XII for the appropriate test condition.
    h. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
    i. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 280 µs from the start of the trace.
    j. The magnitude of the output voltage of item P09 shall satisfy the limits of test test conditions L2 and L6(disregard the polarities of the limits).
    k. Concurrently measure the vertical deflection of item D03 trace, 150 µs from the start of the trace for test conditions L1 and L5, respectively.
    l. The measured value shall satisfy the limits of the test conditions.
21. Test conditions L3, L4, L7, and L8:
    a. Perform test setup per notes 20a - 20f.
    b. Connect the equipment as specified in table XII for the appropriate test condition.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
    d. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 150 µs from the start of the trace for L3 and L7.
    e. The magnitude of the output voltage of item P09 shall satisfy the limits of test conditions L3 and L7(disregard the polarities of the limits).
    f. Concurrently measure the vertical deflection on item D03 trace, 280 µs from the start of the trace for test conditions L4 and L8, respectively.
    g. The measured value shall satisfy the limits of the test conditions.

FIGURE 14. Angular accuracy with chassis noise check test circuit for table XII - Continued.

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22. Test conditions L9 and L11.

    a. Perform test setup per notes 20a - 20f.
    b. Connect the equipment as specified in table XII for appropriate test condition.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
    d. Adjust the output voltage of item P09 for a 0 ±0.5 mV indication on item D03, 150 µs from the start of the trace.
    e. The step between 150 µs and 280 µs from the beginning of the trace shall satisfy the limits of the test conditions.

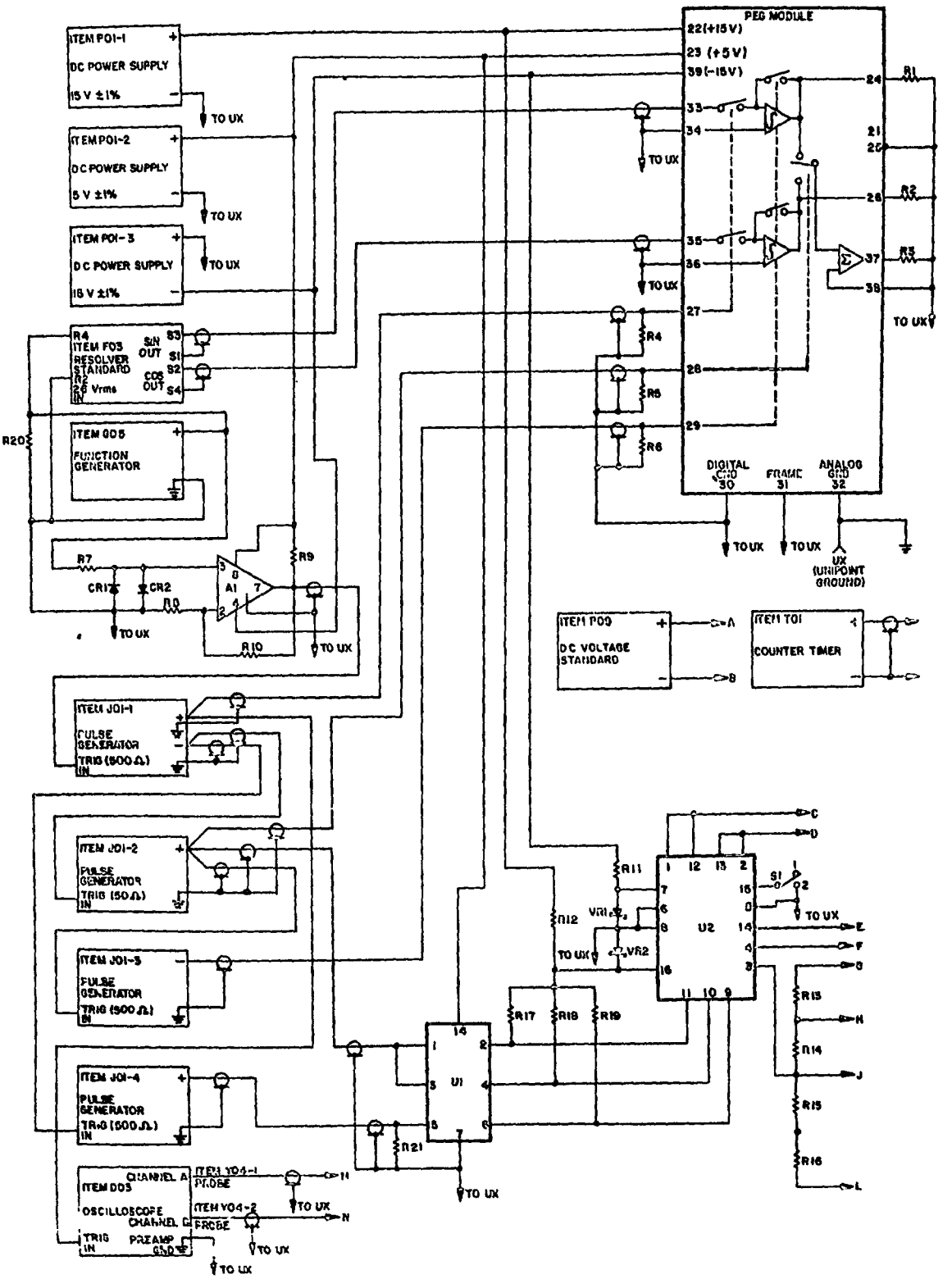
23. Test conditions L10 and L12:

    a. Perform test setup per notes 20a - 20e.
    b. Connect the equipment as specified in table XII for appropriate test condition.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
    d. Set switch S1 to position 2.
    e. Adjust the output voltage of item P09 so that the step between 150 µs and 280 µs from the beginning of the trace is 0 ±0.5 mV.
    f. Simultaneously, measure the average value of the vertical deflection of item D03, 280 µs from the start of the trace.
    g. The value measured in 23f shall satisfy the limits of the test conditions.

24. The vertical amplifier used in item D03 shall be a differential amplifier.

FIGURE 14. Angular accuracy with chassis noise check test circuit for table XII- Continued.

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* FIGURE 15. Harmonic rejection test circuit for table XIII.

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NOTES:
1. ⏚ TO UX = Indicates individual connections that must be made at the unipoint ground (pin 32).
2. All equipment except item D03 must be isolated from earth ground.
3. A1 = LM111.
4. CR1 = CR2 = 1N4148 or equivalent.
5. R1 = R2 = R3 = 15 kΩ ±1%, 1/8 W.
6. R4 = R5 = R6 = R20 = R21 = 51Ω ±5%, 1/2 W.
7. R7 = 10 kΩ ±1%, 1/8 W.
8. R8 = 47CΩ ±1%, 1/8 W.
9. R9 = 300Ω ±5%, 1/4 W.
10. R10 = 47 kΩ ±1%, 1/8 W.
11. R11 = R12 = 750Ω ±5%, 1 W.
12. R13 = 12.5 kΩ ±0.1%, 1/10 W.
13. R14 = R15 = 30 kΩ ±0.05% matched to ±0.02%, 1/10 W.
14. R16 = 45 kΩ ±0.1%, 1/10 W.
15. R17 = R18 = R19 = 20 kΩ ±5%, 1/8 W.
16. S1 = SPDT switch.
17. VR1 = VR2 = 1N4461, or equivalent.
18. U1 = 5417 HEX open-collector buffer.
19. U2 = CD4053B analog switch.
20. Test conditions M1, M2, M5 and M6:
    a. Set item F03 "TYPE" switch to "RESOLVER". Set the E_S(L-L(MAX) "OUTPUT" switch to 26 V and the decade "DEGREES" switches to 0.0000°.
    b. Adjust item G05 for 5.6 ±0.5 V peak triangle wave at 4 400 ±1 Hz at the cosine output terminals of item F03.
    c. Apply the input pulses as specified on figure 10.
    d. Set item D03 trigger to EXT (-) DC, sweep to 50 µs/cm..., and input select to (DC) for both channels.
    e. Set switch S1 to position 1.
    f. Connect the equipment as specified in table XIII for t the appropriate test condition.
    g. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start o of the trace.
    h. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 280 µs from the start of the trace.
    i. The magnitude of the output voltage of item P09 shall satisfy the limits of test conditions M2 and M6 (disregard the polarities of the limits).
    j. Concurrently measure the vertical deflection of item D03 trace, 150 µs from the start of the trace for test conditions M1 and M5.
    k. The measured value shall satisfy the limits of the test conditions.
21. Test conditions M3, M4, M7 and M8:
    a. Perform test set up per notes 20a-20e.
    b. Connect the equipment as specified in table XIII for the appropriate test condition.
    c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
    d. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 150 us from the start of the trace for M3 and M7.
    e. The magnitude of the output voltage of the item P09 shall satisfy the limits of test conditions M3 and M7 (disregard the polarities of the limits).
    f. Concurrently measure the vertical deflection of item D03 trace, 280 µs from the start of the trace for test conditions M4 and M8.
    g. The measured value shall satisfy the limits of the test conditions.
22. Test conditions M9 and M11:

* FIGURE 15. Harmonic rejection test circuit for table XIII - Continued.

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a. Perform test set up per notes 20a-20e.
b. Connect the equipment as specified in table XVIII for appropriate test condition.
c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
d. Adjust the output voltage of item D03, for a 0 ±0.5 mV indication on item P09 150 µsec from the start of the trace.
e. The step between 150 µs and 280 µs from the beginning of the trace shall satisfy the limits of the test conditions.

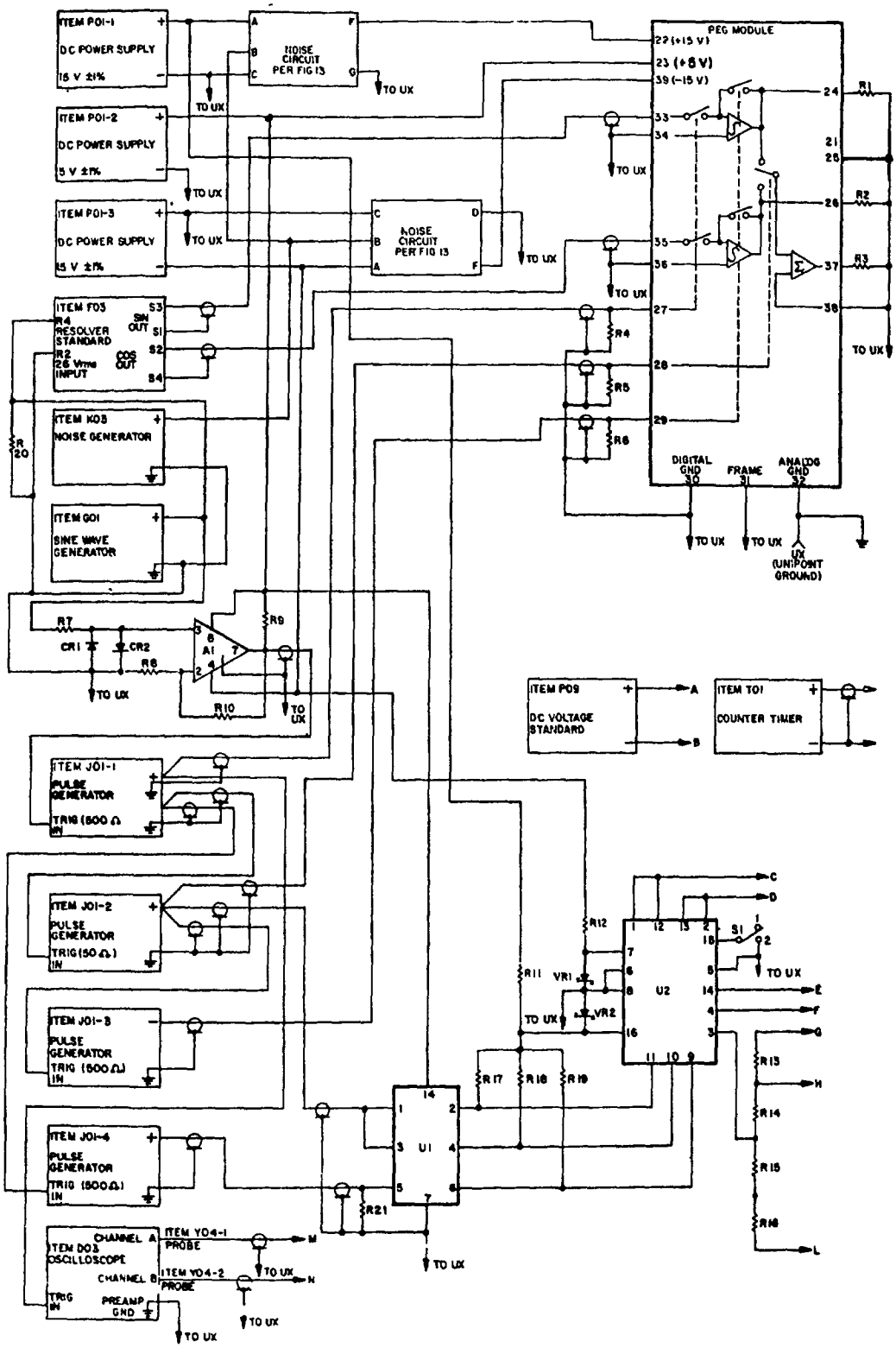
23. Test conditions M10 and M12:

a. Perform test setup per notes 20a-20d.
b. Connect the equipment as specified in table XVIII for appropriate test condition.
c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading 380 µs from the start of the trace.
d. Set switch S1 to position 2.
e. Adjust the output voltage of item P09 so that the step between 150 µs and 280 µs from the beginning of the trace is 0 ±0.5 mV.
f. Simultaneously measure the average value of the vertical deflection of item D03, 280 µs from the start of trace.
g. The value measured in 23f shall satisfy the limits of the test conditions.

24. The vertical amplifier used in item D03 shall be a differential amplifier.

* FIGURE 15. Harmonic rejection test circuit for table XIII - Continued.

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* FIGURE 16. Power supply noise rejection test circuit for table XIV.

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NOTES:
1. ⏚ TO UX = Indicates individual connections that must be made at the unipoint ground (pin 32).
2. All equipment except item D03 must be isolated from earth ground.
3. A1 = LM111.
4. CR1 = CR2 = 1N4148 or equivalent.
5. R1 = R2 = R3 = 15 kΩ ±1%, 1/8 W.
6. R4 = R5 = R6 = R20 = R21 = 51Ω ±5%, 1/2 W.
7. R7 = 10 kΩ ±1%, 1/8 W.
8. R8 = 470Ω ±5%, 1/4 W.
9. R9 = 300Ω ±5%, 1/4 W.
10. R10 = 47 kΩ ±1%, 1/8 W.
11. R11 = R12 = 750Ω ±5%, 1 W.
12. R13 = 12.5 kΩ ±0.1%, 1/10 W.
13. R14 = R15 = 30 kΩ ±0.05% matched to ±0.02%, 1/10 W.
14. R16 = 45 kΩ ±0.1%, 1/10 W.
15. R17 = R18 = R19 = 20 kΩ ±5%, 1/8 W.
16. S1 = SPDT switch.
17. VR1 = VR2 = 1N4461 or equivalent.
18. U1 = 5417 HEX open-collector buffer.
19. U2 = CD4053B analog switch.
20. For test conditions N1-N12, delete the power supply noise circuit connected between item P01-3 and pin 39 of module PEG. Connect the + output of item P01-3 to UX and the - output of item P01-3 to pin 39 of module PEG.
21. For test conditions N13-N24, delete the power supply noise circuit connected between item P01-1 and pin 22 of module PEG. Connect the + output of item P01-1 to pin 22 of module PEG and the - output of item P01-1 to UX.
22. Test conditions (N1, N4, N5, N8-N12) and (N13, N16, N17, N20-N24):
    a. Set item F03 "TYPE" switch to "RESOLVER". Set the E_S(MAX) "OUTPUT" switch to 26 V and the decade "DEGREES" switches to 0.0000°.
    b. Adjust item G01 for 3.535 ±0.035 Vrms, 400 ±1 Hz at the cosine output terminals of item F03.
    c. Adjust K03 for 0.01 ±0.001 Vrms "WHITE NOISE" between the noise test circuit output pin F and UX. Set the item K03 bandwidth to 5 Hz to 20 kHz.
    d. Apply the input pulses as specified on figure 10.
    e. Set item D03 trigger to EXT(-)DC, sweep to 50 µs/cm, and input select to (DC) for both channels.
    f. Set switch S1 to position 1.
    g. Connect the equipment as specified in table XIV for the appropriate test condition.
    h. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
    i. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 280 µs from the start of the trace.
    j. The magnitude of the output voltage of item P09 shall satisfy the limits of test conditions N2 or (N14) and N6 or (N18)(disregard the polarities of the limits).
    k. Concurrently measure the vertical deflection of item D03 trace, 150 µs from the start of the trace for test conditions N1 or (N13) and N5 or (N17), respectively.
    l. The measured value shall satisfy the limits of the test conditions.

* FIGURE 16. Power supply noise rejection test circuit for table XIV - Continued.

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23. Test conditions N3, N4, N7 and N8 and (N15, N16, N19 and N20):

a. Perform test setup per notes 22a - 22f.
b. Connect the equipment as specified in table XIV for the appropriate test condition.
c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
d. Adjust the output voltage of item P09 for a 0 ±2 mV indication on item D03, 150 µs from the start of the trace for N3 or (N15) and N7 or (N19).
e. The magnatude of the output voltage of item P09 shall satisfy the limits of table I for test conditions N3 or (N15) and N7 or (N19)(disregard the polarities of the limits).
f. Concurrently measure the vertical deflection of item D03 trace, 280 µs from the start of the trace for test conditions N4 or (N16) and N8 or (N20), respectively.
g. The measured value shall satisfy the limits of the test conditions.

24. Test conditions N9 and N11 and (N21 and N23):

a. Perform test setup per notes 22a - 22f.
b. Connect the equipment as specified in table XIV for appropriate test condition.
c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
d. Adjust the output voltage of item P09 for a 0 ±0.5 mV indication on item D03, 150 µs from the start of the trace.
e. The step between 150 µs and 280 µs from the beginning of the trace shall satisfy the limits of test conditions.

25. Test conditions N10 and N12 and (N22 and N24):

a. Perform test setup per notes 22a - 22e.
b. Connect the equipment as specified in table XIV for appropriate test condition.
c. Set the vertical sensitivity of item D03 to 1 mV/cm. Adjust the vertical position for a 0 volt reading, 380 µs from the start of the trace.
d. Set switch S1 position 2.
e. Adjust the output voltage of item P09 so that the step between 150 µs and 280 µs from the beginning of the trace is 0 ±0.5 mV.
f. Simultaneously, measure the average value of the vertical deflection of item D03 280 µs from the start of the trace.
g. The value measured in 25f shall satisfy the limits of the test conditions.

26. The vertical amplifier used in item D03 shall be a differential amplifier.

* FIGURE 16. Power supply noise rejection test circuit for table XIV - Continued.

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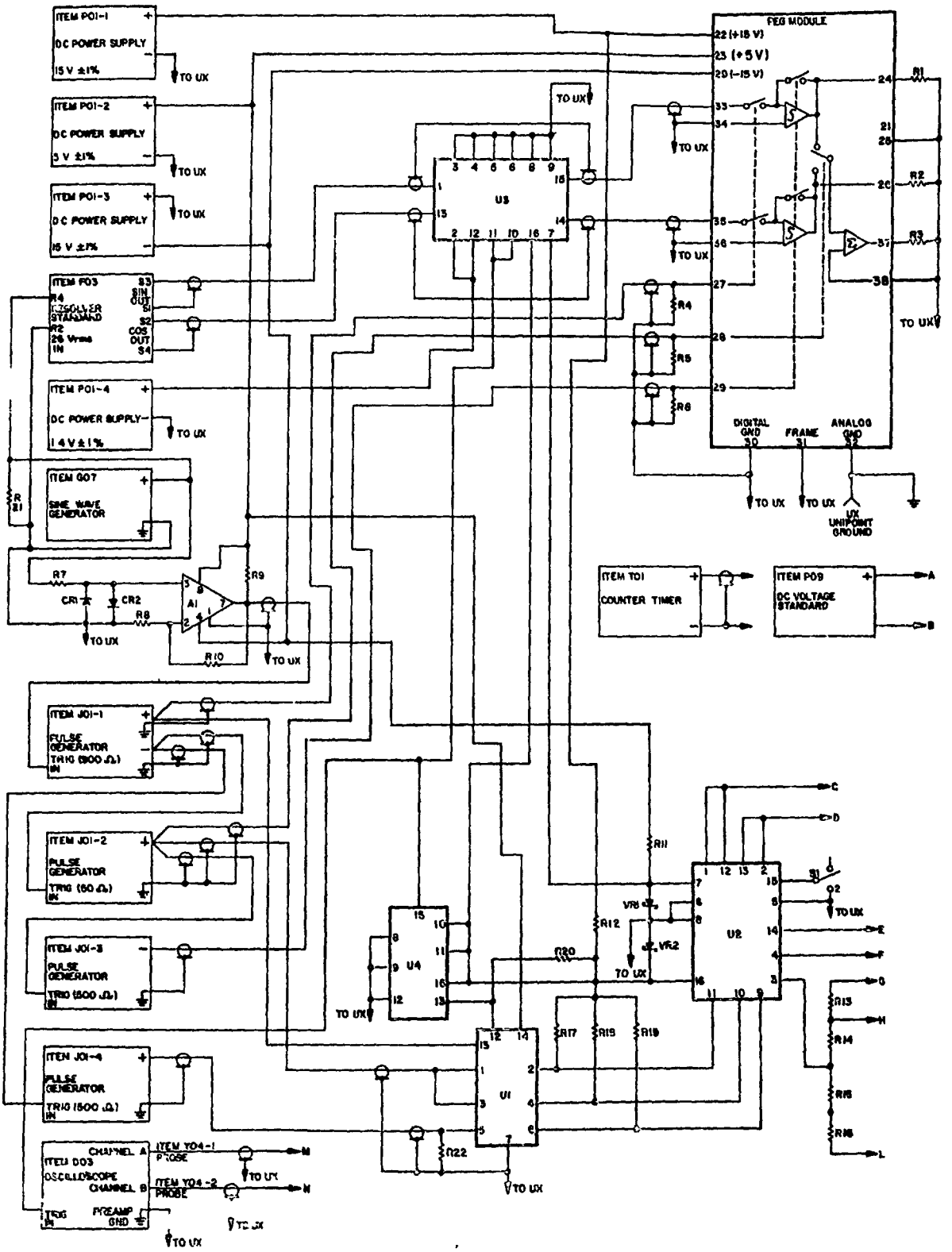


FIGURE 17. Angular accuracy repeatibility check test circuit for table XV.

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NOTES:
1. ⏚ TO UX = Indicates individual connections that must be made at the unipoint ground (pin 32).
2. All equipment except item D03 must be isolated from earth ground.
3. A1 = LM111.
4. CR1 = CR2 = CR2 = 1N4148 or equivalent.
5. R1 = R2 = R3 = 15 kΩ ±1%, 1/8 W.
6. R4 = R5 = R6 = R21 = R22 = 51Ω ±5%, 1/2 W.
7. R7 - 10 kΩ ±1%, 1/8 W
8. R8 = 470Ω ±1%, 1/8 W.
9. R 9 = 300Ω ±5%, 1/4 W.
10. R10 = 47 kΩ ±1%, 1/8 W.
11. R11 = R12 = 750Ω ±5%, 1 W.
12. R13 = 12.5 kΩ ±0.1%, 1/10 W.
13. R14 = R15 = 30 kΩ ±0.05% matched to ±0.02%, 1/10 W.
14. R16 = 45 kΩ ±0.1%, 1/10 W.
15. R17 = R18 = R19 = R20 = 20 kΩ ±5%, 1/8 W.
16. S1 = SPDT switch.
17. VR1 = VR2 = 1N4461, or equivalent.
18. U1 = 5417 HEX open-collector buffer.
19. U2 = U3 = CD4053B analog switch.
20. U4 = CD4027B dual J-K master-slave flip-flop.
21. Test conditions P1 and P4:

    a. Set item F03 "TYPE" switch to "RESOLVER". Set the E_S(MAX) "OUTPUT" switch to 26 V and the decade "DEGREES" switch to 0.0000°.
    b. Adjust item G07 for 3.535 0.035 Vrms, 400 ±1 Hz between pin 13 of U3 and UX.
    c. Apply the input pulses as specified on figure 10, except that PWC shall be 1.0 ±0.02 ms.
    d. Set item D03 trigger to EXT (+) DC, sweep to 1 ms/cm., and input select to (DC) for both channels.
    e. Set the horizontal display to the 5X position and adjust the horizontal position control to position the start of the trace at the left side of the graticule.
    f. Set switch 1 to position 1.
    g. Connect the equipment as specified in table XV for the appropriate test condition.
    h. Set item D03 vertical sensitivity to 1 mV/cm.
    i. Measure the vertical deflection of item D03 trace, 1250 µs from the beginning of the trace for P1 and 1350 µs for P4. Set the horizontal display to the 1X position so that several functional cycles are observed. Measure the vertical deflection of the trace at 1.25 ms and 6.25 ms for P1 and 1.35 and 6.35 ms for P4.
    j. The measured difference shall satisfy the limits of the test conditions.

22. Test conditions P2 and P3:

    a. Repeat steps 21a through 21h.
    b. Adjust the output voltage of item P09 for 0 ±2 mV indication on item D03, 1250 µs from the start of the trace for P3 and 1350 µs for P2. Set the horizontal display to the 1X position so that several functional cycles are observed, Measure the vertical deflection of the trace at 1.25 ms and 6.25 ms for P3 and 1.35 ms and 6.35 ms for P2.
    c. The measured differences shall satisfy the limits of the test conditions.

23. Test conditions P5 and P6:

    a. Repeat steps 21a through 21h.
    b. Set switch S1 to position 2.
    c. Set the vertical sensitivity of item D03 to 1mV/cm.
    d. Adjust the output of the item P09 for 0 ±0.5 mV indication on item D03, 1250 µs from the start of the trace. Measure the height of the step in the trace at about 1280 µs. Set the horizontal display to the 1X position so that several functional cycles are observed. Measure the height of the step in the trace at 1.28 ms and at 6.28 ms.
    e. The difference between the measured readings shall satisfy the limits of the test conditions.

24. The vertical amplifier used in item D03 shall be a differential amplifier.

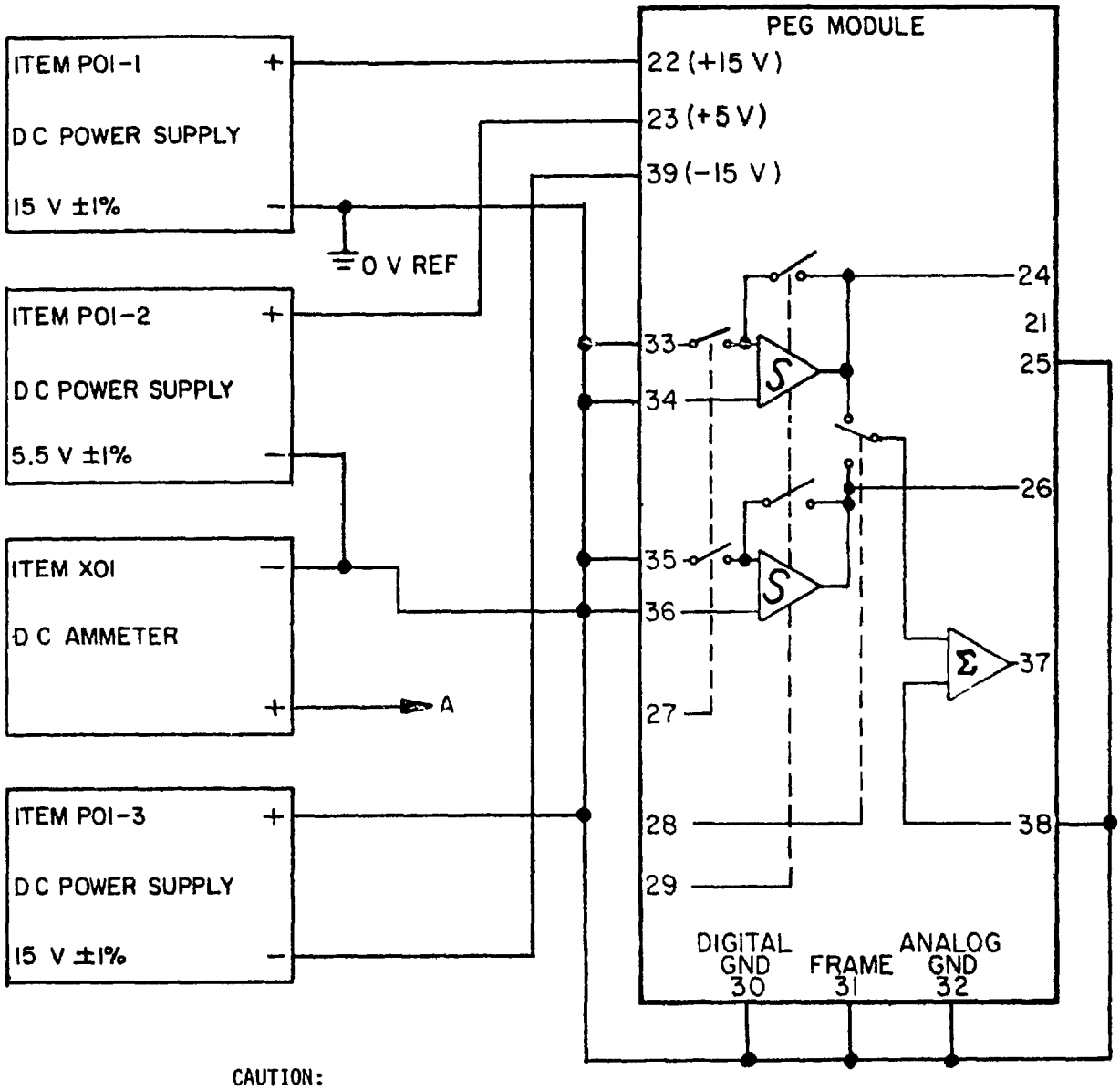
FIGURE 17. Angular accuracy repeatability check test circuit for table XV - Continued.

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TABLE XVI. Input current (low-level) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	Input current, low level	27	0.40	1.73	mA	18
Q2	Input current, low level	28	0.40	1.73	mA	18
Q3	Input current, low level	29	0.40	1.73	mA	18

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTE: Wait 1 minute before taking the measurement.

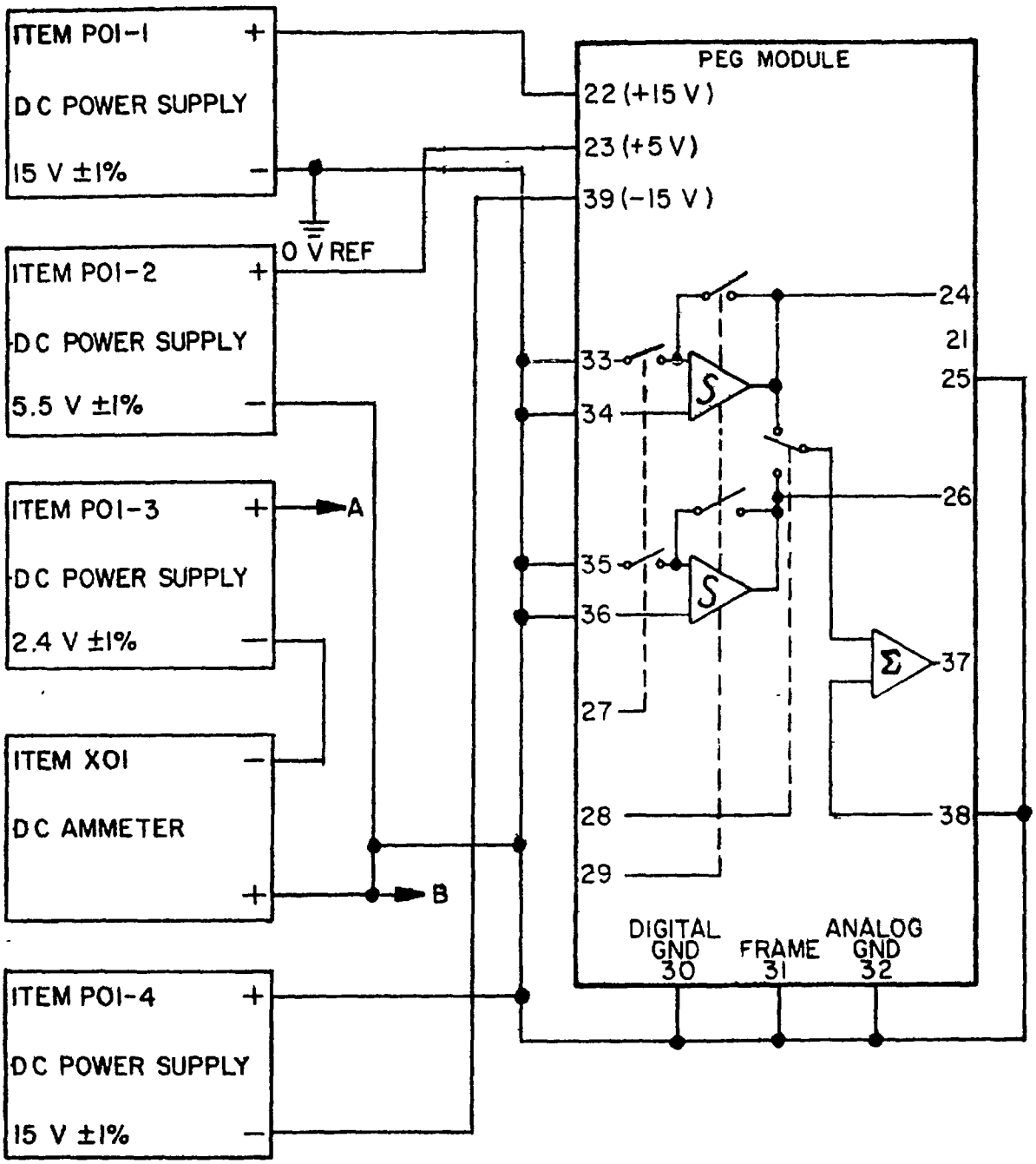
FIGURE 18. Input current (low-level) test circuit for table XVI.

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* TABLE XVII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Input current, high level	27	28,29	0	40	µA	19
R2	Input current, high level	28	27,29	0	40	µA	19
R3	Input current, high level	29	27,28	0	40	µA	19

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

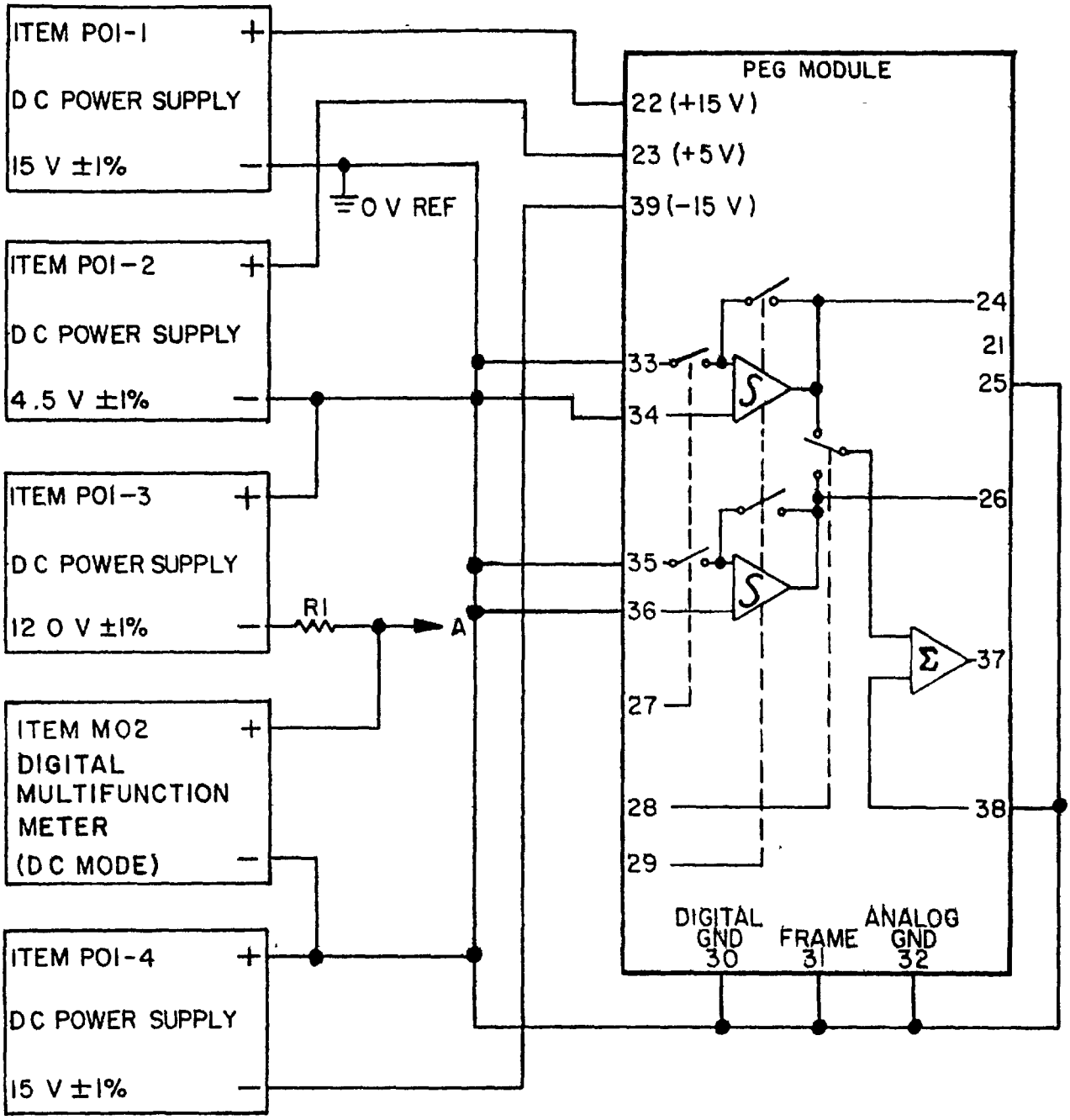
FIGURE 19. Input current (high-level) test circuit for table XVII.

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TABLE XVIII. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
S1	Input clamp diode voltage	27	-0.4	-1.5	V	20
S2	Input clamp diode voltage	28	-0.4	-1.5	V	20
S3	Input clamp diode voltage	29	-0.4	-1.5	V	20

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NOTE: R1 = 875Ω ±1%, 1/4 W.

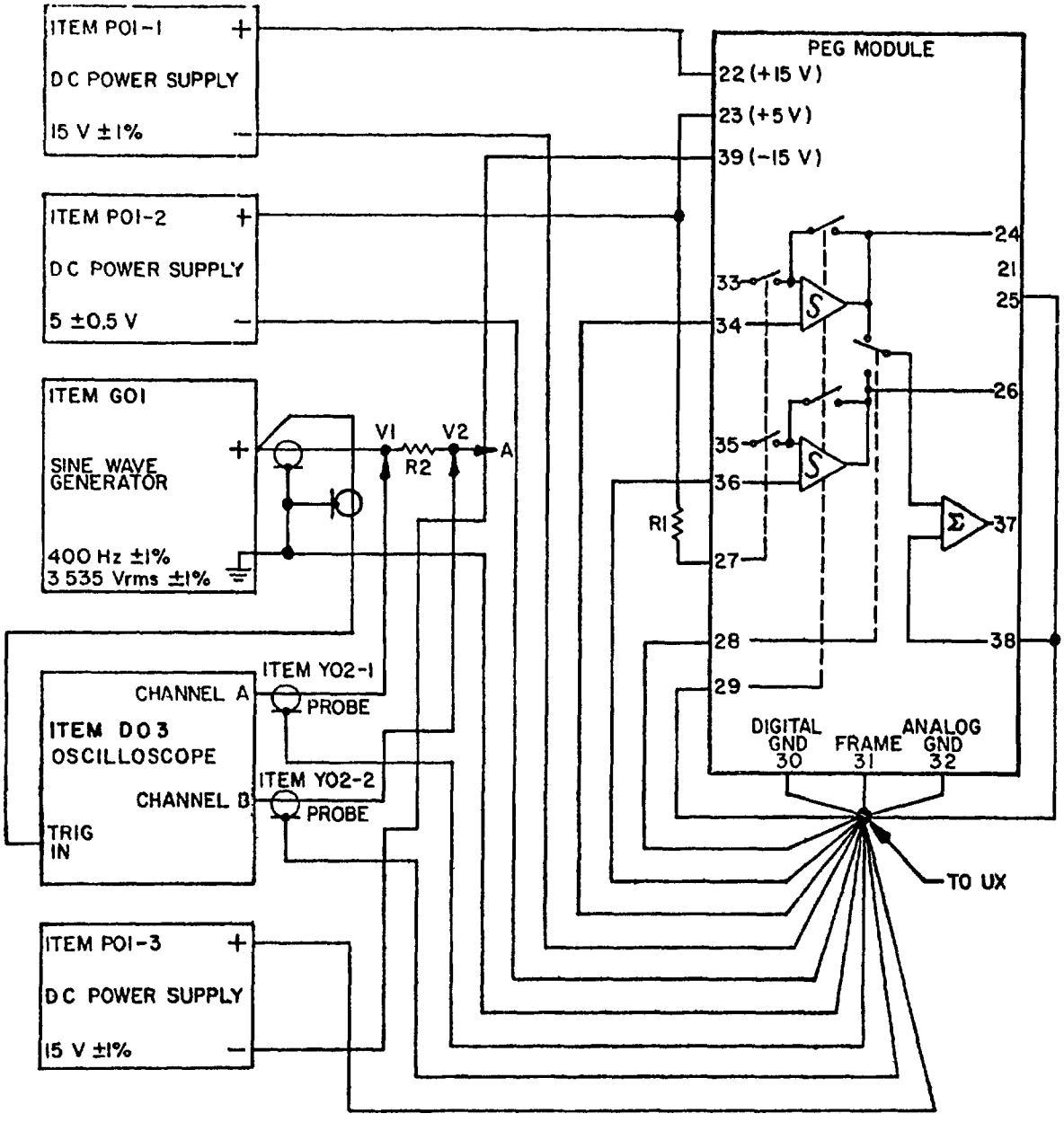
FIGURE 20. Input clamp diode voltage test circuit for table XVIII.

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TABLE XIX. Analog input impedance check test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
T1	Channel A Analog input impedance check	33	0.85	2.40	V_{p-p}	21
T2	Channel B Analog input impedance check	35	0.85	2.40	V_{p-p}	21

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NOTES:
1. R1 = 1 kΩ ±1%, 1/8 W.
2. R2 = 1 MΩ ±1%, 1/8 W.
3. Measure V1-V2 using A-B setting on item D03 preamp.
4. UX = Unipoint ground.
5. All equipment except item D03 shall be isolated from earth ground.
6. The vertical amplifier used in item D03 shall be a differential amplifier.

FIGURE 21. Analog input impedance check test circuit for table XIX.

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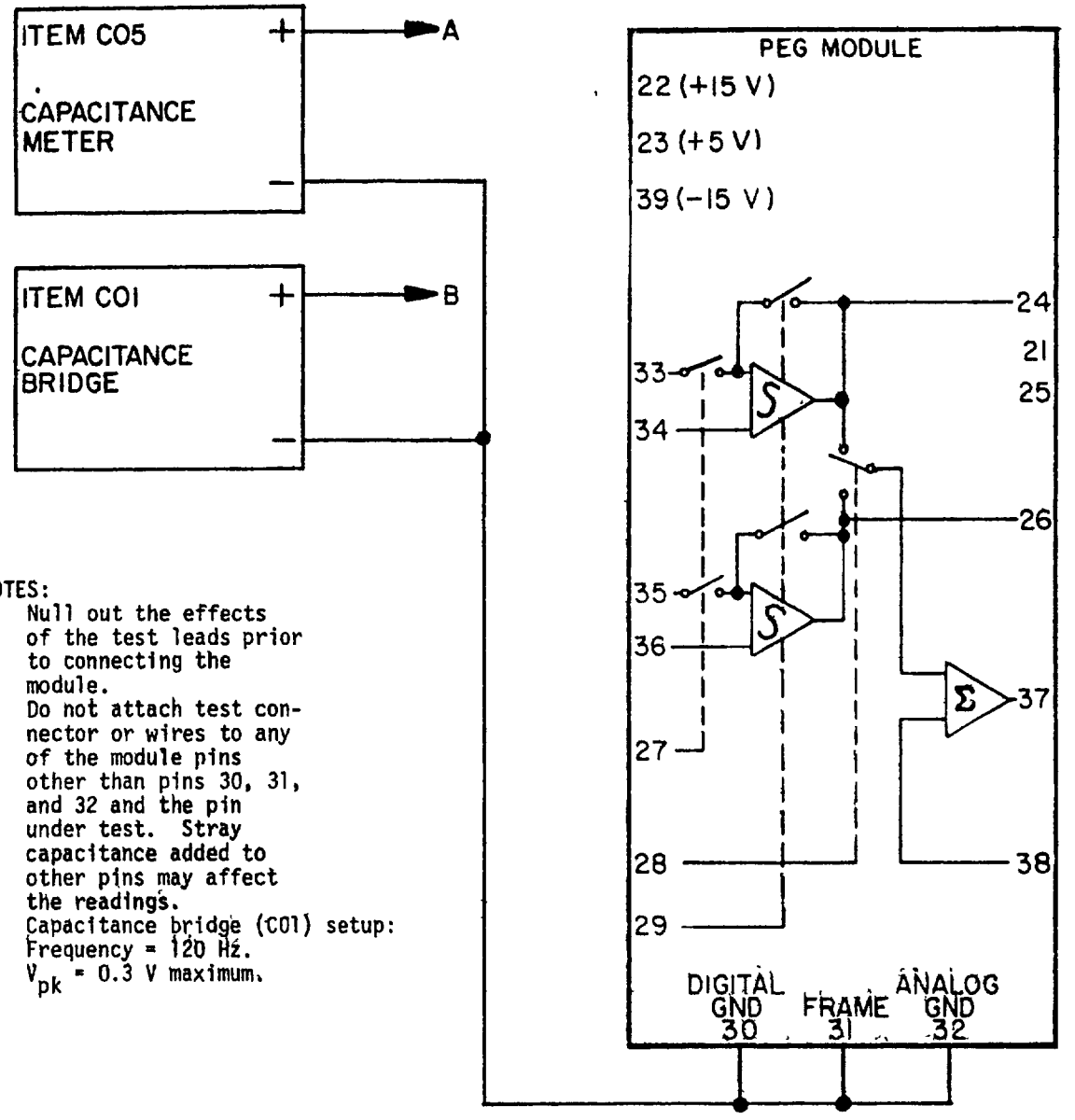
* TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	Input capacitance	27	Open	2	15	pF	22
U2	Input capacitance	28	Open	2	15	pF	22
U3	Input capacitance	29	Open	2	15	pF	22

* TABLE XXI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	Power supply filter capacitance	Open	22	0.03	0.22	µF	22
V2	Power supply filter capacitance	Open	23	0.08	0.22	µF	22
V3	Power supply filter capacitance	Open	39	0.08	0.22	µF	22

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 30, 31, and 32 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (CO1) setup: Frequency = 120 Hz. V_{pk} = 0.3 V maximum.

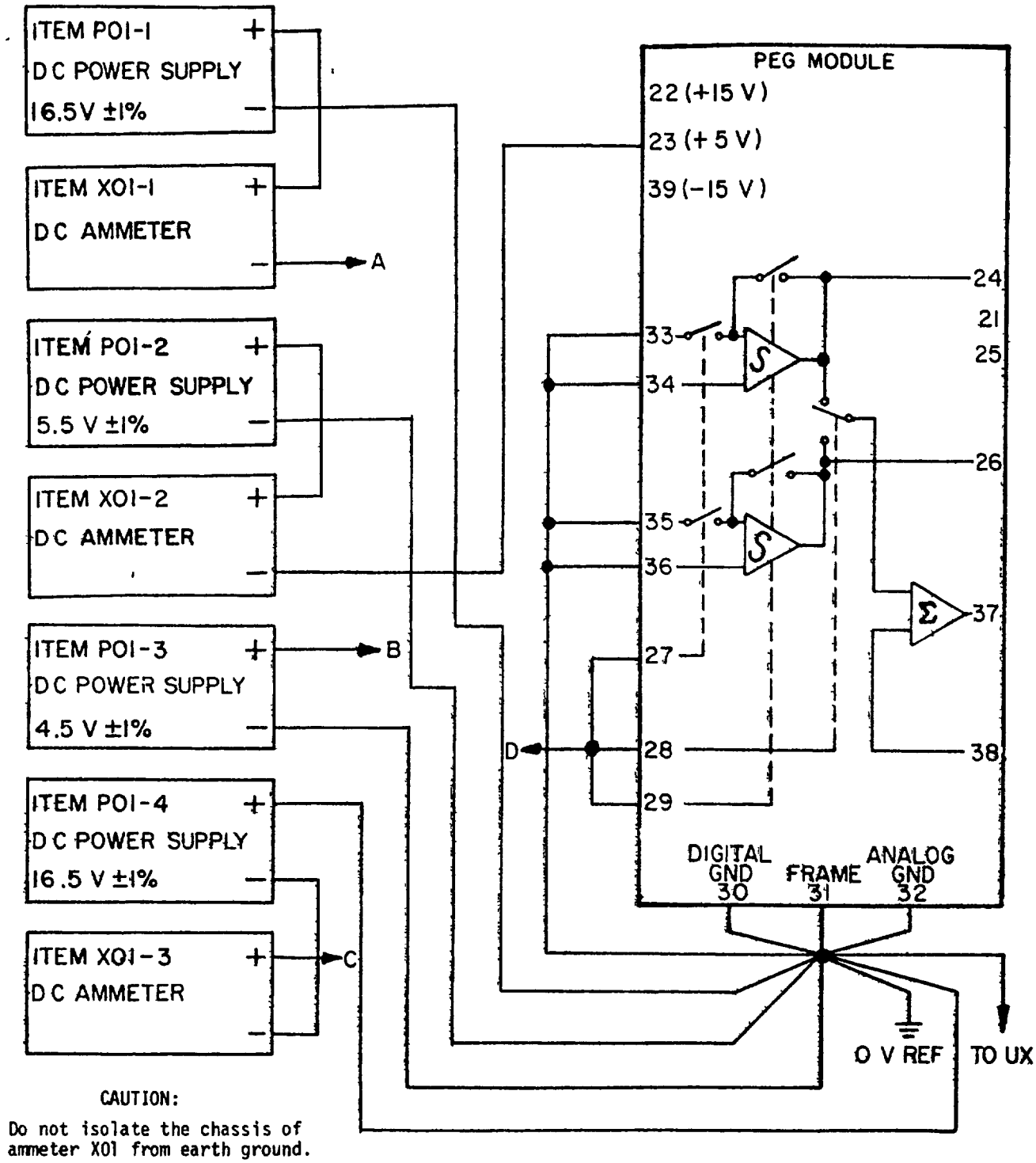
FIGURE 22. Input and power supply filter capacitance test circuit for tables XX and XXI.

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* TABLE XXII. Power supply current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	C to pin	D to pin	Min	Max		
W1	Power supply current, high level	22	39	B	5	41	mA	23
W2	Power supply current, low level	↓	↓	UX	5.0	35.5	↓	↓
W3	Power supply current			UX	7.3	12.7		
W4	Power supply current			UX	7.3	12.7		

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CAUTION:
Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since ammeters X01-1, X01-2, and X01-3 are not used simultaneously only one ammeter is required.
2. For test conditions W1 and W2 read ammeter X01-2.
3. For test condition W3 read ammeter X01-1.
4. For test condition W4 read ammeter X01-3.

FIGURE 23. Power supply current test circuit for table XXII.

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TABLE XXIII. Analog output impedance check test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
X	Analog output impedance check	0.99910	1.00000	V/V	24

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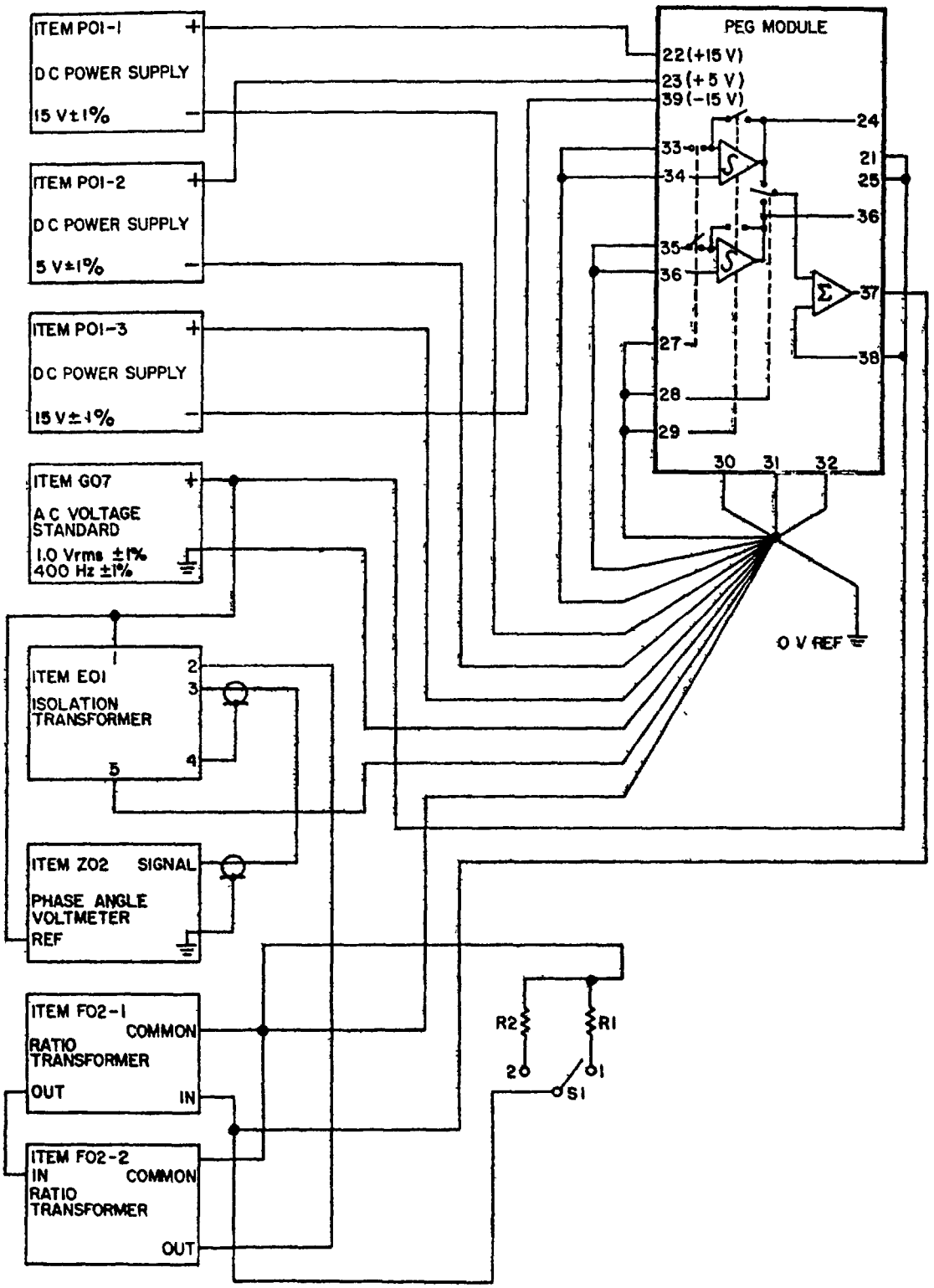


FIGURE 24. Analog output impedance check test circuit for table XXIII.

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NOTES:
1. R1 = 2 kΩ ±1%, 1/8 W.
   R2 = 10 kΩ ±1%, 1/8 W.
2. Set switch S1 in position 1.
3. Adjust item G07 for 5.0 Vrms ±1%, 400 Hz ±1% at pin 38.
4. Set item F02-2 to 1.000000.
5. With the function switch of item Z02, set to 0°, adjust item F02-1 for a null reading on item Z02.
6. Set switch S1 to position 2.
7. Adjust item F02-2 for a null reading on item Z02.
8. The setting of item F02-2 shall satisfy the limits of the test condition.

FIGURE 24. Analog output impedance check test circuit for table XXIII - Continued.

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